UNIT-1

↳ Chapters from TB -

(1), (2), (6)

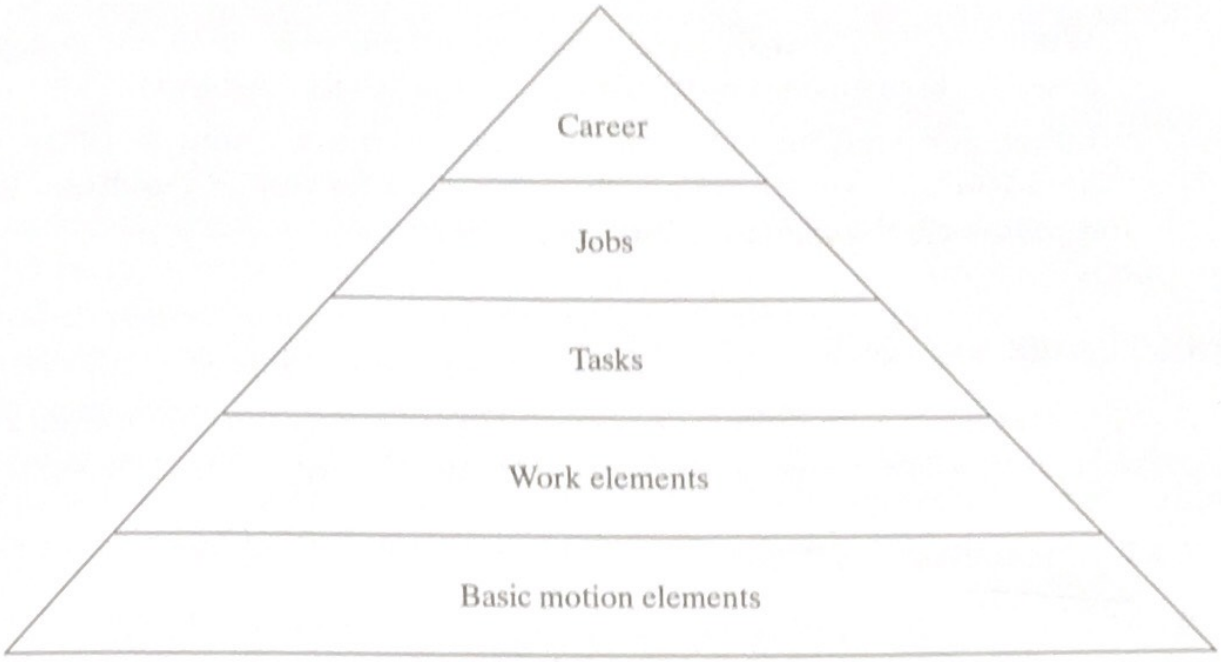


Figure 2 The pyramidal structure of work.

UNIT - 1

Q. What is WORK?

Ans- **Work**: It is defined as an activity in which a person exerts physical, mental & congential efforts to accomplish a given task.

- Work consists of many tasks.
- **Task**: It is the amount of work assigned to a worker.

↳ **Productivity** -

Def: It is the measure of production in relation with its I/P & O/P.

ie $$\text{Productivity} = \frac{O/P}{I/P}$$

↳ Inputs can be -
- Efforts
- Money
- Materials
- Energy

↳ Problems encounted while measuring productivity -

(1) **Non-Homogenous Output Units** - Here the O/P unit may differ in model, price, features etc

(2) **Product Mix Changes** - Here the I/P & O/P are different from product to product.

- Product Mix can also occur in the same category of products.

eg- Different kinds of Rice, in Rice Category.

③ Price & Cost Changes -

Here Price - Refers to the Selling Price

Cost - Refers to the costs incurred during manufacturing

ie Price = Cost + Taxes + Supply / Demand + Logistics Surges

&

Cost = Materials + Labour + Variances

④ Multiple Input Factors -

- As an alternative to measure productivity - Labour Productivity Index (LPI) is measured.

↳ Labour Productivity Ratio (LPR)

$$LPR = \frac{O/P \text{ units}}{\text{Worker Hours}} = \frac{WO}{LH} = \frac{\text{Work Units}}{\text{Labour Hours}}$$

SI unit is = Units / LH

- Here productivity is a fn of O/P & I/P.

- Base Period Productivity : It is the productivity of any year in the past under consideration ie (LPR_b)

- Current Period Productivity : It is the productivity of the present year. ie (LPR_t)

∴ Labour Productivity Index is -

$$LPI = \frac{LPR_t}{LPR_b}$$

If $LPI = 1$, then Productivity b/w Base & Current Period is SAME.

If $LPI > 1$, then Productivity b/w Base & Current Period has ↑ (INCREASED)

If $LPI < 1$, then Productivity b/w Base & Current Period has ↓ (DECREASED)

↳ Case ① - A work group of 5 workers in a certain month produced 500 units of o/p, working 8 hrs per day for 22 days in a month. Suppose in the next month, the same group produced 600 units but this was done only in 20 days in a month. Determine the productivity index, using the prior month as a Base.

Note - If no. of working hrs are NOT mentioned then take it as = 8 hrs = 1 shift

Sol- (1) ~~Base Group~~:

Base Period:

No. of Worker = 5

o/p units = 500

Dur = 8hrs/day for 22 days

$\therefore LPR_b = \frac{500}{8 \times 5 \times 22}$

$\therefore LPR_b = 0.568$ units/LH

Current Period:

No. of worker = 5

o/p units = 600

Dur = 8hrs/day for 20 days

$\therefore LPR_t = \frac{600}{5 \times 8 \times 20}$

$\therefore LPR_t = 0.75$ units/LH

$\therefore$ WRT- $LPI = \frac{LPR_t}{LPR_b}$

$\therefore LPI = \frac{0.75}{0.568} = 1.32$

Here $LPI > 1$

$\therefore$ Change in Productivity is $(1.32 - 1) \times 100$

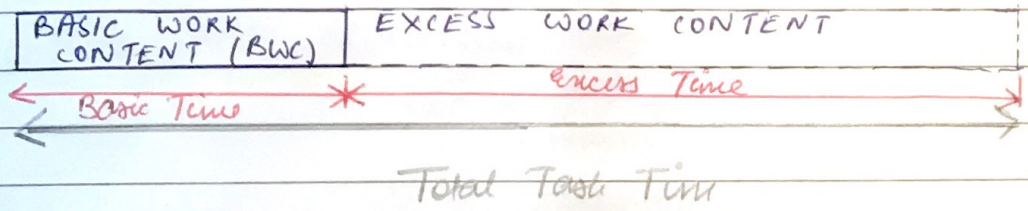
$= 32\%$

$\therefore$ Productivity increased by 32% from the base period to the current period.

- For Labour Hour = No. of workers × No. of Hours × No. of days

- For Labour Month = No. of Workers × No. of days

⇒ Productive Work Content -



- Basic Work Content (BWC) - It is a theoretical minimum amount of work req. to achieve a task.
  - Here the amt of work req is expressed in terms of Time.
- Basic time :- It is the IRREDUCABLE amount of Time req to achieve a task.
- Excess Work Content - It is the amount of work expressed in time, that adds & facilitates no added value to the product.
- It can be caused due to — (Causes of Excess Work Content).
  - (a) Poor design of Product
  - (b) Poor Workout Layout
  - (c) Due to Human factor.

4.2 Productive Work Content

Work is usually not performed in the most efficient way possible. A given task performed by a worker can be considered to consist of (1) the basic productive work content and (2) excess nonproductive activities. In general, the task is associated with the production of a product or the delivery of a service. The ***basic productive work content*** is the theoretical minimum amount of work required to accomplish the task, where the amount of work is expressed in terms of time. Thus, the time required to perform the basic productive work content is the theoretical minimum. It cannot be further reduced. The basic productive work content time is a theoretical concept implying optimal conditions that rarely, if ever, are achieved in practice. Nevertheless, it is an optimum worth seeking.

The ***excess nonproductive activities*** in the task are the extra physical and mental actions performed by the worker that do not add any value to the task, nor do they facilitate the productive work content that does add value. The excess nonproductive activities take time. They add to the basic productive work content time to make up the total time needed to perform the task. The excess nonproductive activities can be classified into three categories, as illustrated in the time line in Figure 4:

- Excess activities caused by poor design of the product or service
- Excess activities caused by inefficient methods, poor work layout, and interruptions
- Excess activities caused by the human factor

Some examples of the three categories of nonproductive activities are listed in Table 8. As several of these examples indicate, nonproductive activities do not necessarily occur during every cycle of the task. For example, industrial accidents occur infrequently in most work situations. However, over the long run, their effects extract a heavy toll on productive work content.

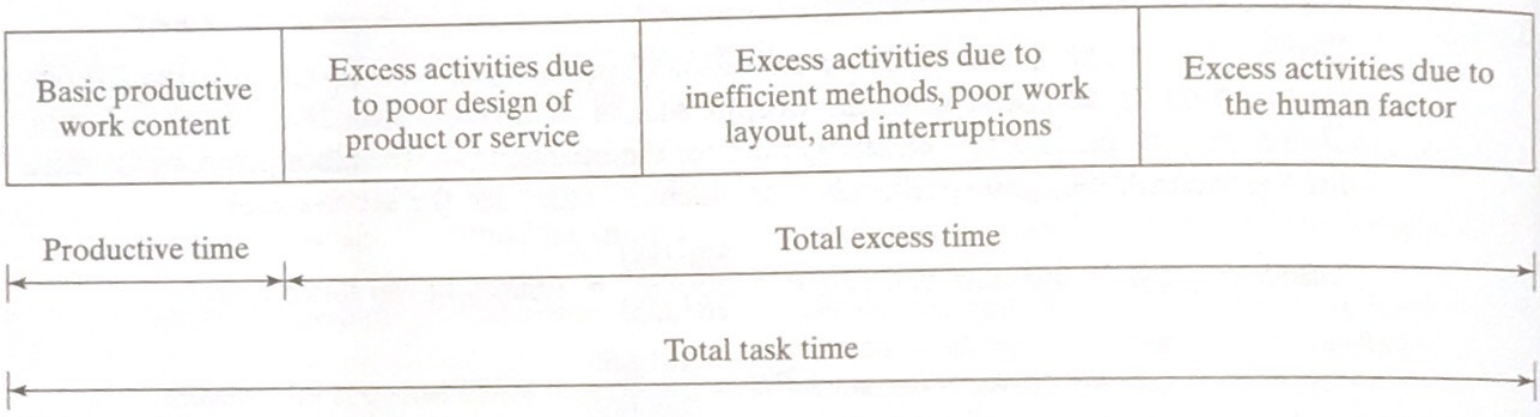


Figure 4 Total task time consists of basic productive work content and nonproductive activities.

- A designer working at a computer-aided design (CAD) station to design a new product
- A construction project consisting of a work crew building a highway bridge

As our list indicates, a work system can include one or more human workers. It can also include automated systems that operate for extended periods of time without human attention. Sooner or later, automated systems require the attention of human workers for purposes of maintenance or reprogramming or other reasons. The information associated with the work system may consist simply of a worker's knowledge required to perform an operation, or it may involve the use of databases and programs that must be accessed using computer systems. The equipment in a work system may be a single machine tool operated by one worker or a collection of automated machines that operate under computer control in a coordinated fashion.

2.2 Work Systems as a Field of Professional Practice

As a field of professional practice, ***work systems*** includes (1) work methods, (2) work measurement, and (3) work management.[5] The term ***work science*** is often used for this professional practice. The field of ***work methods*** consists of the analysis and design of tasks and jobs involving human work activity. Terms related to work methods include ***operations analysis*** and ***methods engineering***. The term ***motion study*** is also used, but its scope is usually limited to the physical motions, tools, and workplace layout used by a worker to perform a task. The other terms are less restrictive and include the analysis and design of complex processes consisting of material and information flows through multiple operations.

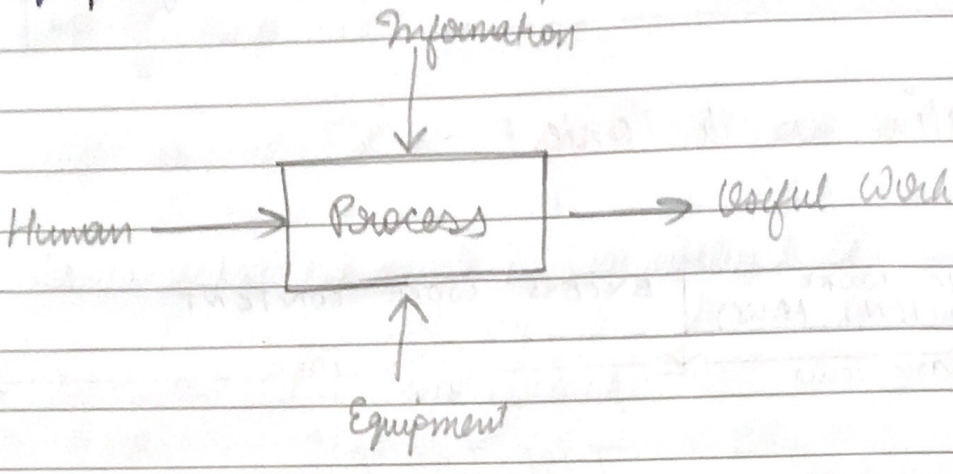
Work measurement is the analysis of a task to determine the time that should be allowed for a qualified worker to perform the task. The time thus determined is called the ***standard time***. Among its many applications, the standard time can be used to compute product costs, assess worker performance, and determine worker requirements (e.g., how many workers are needed to accomplish a given workload). Because of its emphasis on time, work measurement is often referred to as time study. However, in the modern context, ***time study*** has a broader meaning that includes work situations in which an operation is performed by automated equipment, and it is desired to determine the cycle time for the operation. Thus, time study covers any and all work situations in which it is necessary to determine how long it takes to accomplish a given unit of work, whether the work unit is concerned with the production of a product or the delivery of a service. Time is important because it equates to money ("time is money," as the saying goes), and money is a limited resource that must be well managed in any organization.

Work management refers to the various organizational and administrative functions that must be accomplished to achieve high productivity of the work system and effective supervision of workers. Work management includes functions such as (1) organizing workers to perform the specialized tasks that constitute the workload in each

[5]The title of this text emphasizes the two definitions: (1) work systems as physical entities and (2) work systems as a field of professional practice.

→ WORK SYSTEMS - (As a Physical Entity)

Def: It is a system consisting of Human, Information & Equipment that is designed to perform useful work.



- It is of 3 types -

① Manual Work Systems

② Hybrid Work Systems

③ Automated Work Systems.

↳ ① Manual Work System -

- It consists of a worker performing 1 or more tasks without any "POWERED TOOLS"
- It incorporates the use of basic tools or HAND tools.
- There is no external source of power.
- There are 2 types of manual Work Systems -

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(a) Pure Manual Worksystems.

(b) Manual Work using Hand Tools.

- Manual Work can be categorised into 2 time cycles -

(a) Repeative - Here the work cycle is short & it has High Degree of similarity.

(b) Non-Repeatative: Here the work cycle is longer & the successive cycles are not similar.

- NOTE - MANUFACTING LEAD TIME is always SMALLER than SERVICE LEAD TIME.

- The lead time consists of many work cycles that are calculated as a fn of Time.

- NORMAL TIME (T_n) - It is defined as the time taken when a 'WORK CYCLE' is performed at 100% Performance. aka "Basic Time."

- STANDARD TIME (T_{std}) - It is defined as the time, which is inclusive of all the breaks provided throughout the shift to workers with respect to 1 unit.

- CYCLE TIME (T_c) - It is the time elapsed to produce 1 unit of product.

Here $$T_c = \frac{T_n}{P_w}$$ → Worker's Performance.

- Cycle Time is Σ (Work Elements).

- Performance - It is the pace at which a task is carried out.

Q. What is Standard Method?

Ans- Standard Method - It is the method of work carried out by skilled workers with Standard Performance.

ie Standard Performance $\propto \frac{1}{\text{Cycle Time}}$

Q. What are Allowances?

Ans- Allowances - It is the time given to tackle the variability of a worker & it accomodates, lunch breaks, tea breaks etc.

- Allowances are dependent on the amount of work.
- Primarily they are of 3 types - ie Apfd

P → Personal Allowances

f → Fatigue Allowances

d → Delay Allowances.

↳ Relationship b/w Normal, Standard Times & Allowances -

$$T_{std} = T_n \left(1 + A_{pfd}\right)$$

NOTE - IRREGULAR Work Elements are accounted for and added & considered in the Normal Time in the frequency of their appearances.

↳ Relationship b/w Shift Time, T_{std} & Units Produced (φ).

$$H_{sh} = \varphi \cdot T_{std}$$

H_{sh} → Shift Time

↳ Relationship b/w φ_{std}, Shift Time & T_{std}.

$$\varphi_{std} = \frac{\text{Shift Time}}{T_{std}}$$

Here φ_{std} = Standard Quantity.

↳ Actual Time Worked -

$$\text{Shift Time} = \frac{\text{Actual Time Worked}}{\left(1 + A_{pfd}\right)}$$

↳ Lost Time -

$$\text{Lost Time} = \text{Shift Time} - \text{Actual Time Worked}$$

↳ Daily Production Rate (R_p)

$$R_p = \frac{\text{Actual Time Worked}}{T_c}$$

Q. How to measure a worker's Productivity?

Ans- The manual worker's productivity is measured by -

① Standard Hours (H_{std})

② Worker Effeciency (E_w)

ie ① Standard Hours -

$$H_{std} = Q \cdot T_{std}$$

② Worker's Effeciency -

$$E_w = \frac{H_{std}\ \%}{H_{sh}}$$ It is expressed as a %.

Q. What are the rubricks used to measure a workers performance & analyse how these rubricks are influenced by the actual no. of hours worked?

Ans- ie $$Q_{actual} = \frac{\text{Actual Time Worked}}{T_c}$$

- Actual Time Worked (aka Actual no. of Hours worked) influence the Actual no. of units produced.
- Then Standards Hours (H_{std}) is the 1st Rubrick -

$$H_{std}\ \text{Actual} = \text{Actual} \times T_{std}$$

& Hence the no. of hours worked Influences Standard Hours.

- Then Worker's Efficiency (E_w) is the 2nd Rubrick -

$$E_w = \frac{H_{std}}{H_{sh}}$$

∵ H_{std} is already influenced by the no. of hours worked, then even E_w is said to influenced by the no. of hours worked.

↳ Case -

(a) The normal time of a particular activity is 3.23 mins & the irregular work element, whose normal time is 1.25 mins is performed every 5 cycles. The allowance are 15% Determine the Standard Time.

Sol- Irregular Work Element → 5 cycles → $\frac{3.23 \text{ mins}}{1.25}$

∴ For 1 cycle → $\frac{1.25}{5} = 0.25$ mins

∴ $T_n = 3.23 + 0.25 = 3.48$ mins

∴ $T_n = 3.48$ mins

Here $A_{pfd} = 15\% = \frac{15}{100}$

WKT-

$$T_{std} = T_n (1 + A_{pfd})$$

$$T_{std} = 3.48 \left(1 + \frac{15}{100}\right)$$

$$\therefore T_{std} = 4 \text{ mins}$$

(b) Continuing, how many work units are produced if we consider the worker's performance in an 8hr shift is standard.

Sol-

Here Q_{std} = Standard quantity.

$$\therefore Q_{std} = \frac{H_{sh}}{T_{std}}$$

$$Q_{std} = \frac{8 \times 60}{4}$$

$$Q_{std} = 120 \text{ units}$$

(c) Continuing, also find the time lost due to the allowance factors.

Sol-

Here -

Shift Time = Actual Time Worked (1 + Apfd)

$\therefore$ Actual Time Worked = $\frac{\text{Shift Time}}{1 + Apfd}$

$\therefore$ Actual Time Worked = 6.95 hours

$\therefore$ Lost Time = Shift Time − Actual Time Worked

$\therefore$ Lost Time = 1.05 Hours

(d) Continuing, The production rate when the worker's performance exceeds 100%, assuming that the worker is performing at 125%, then determine the no. of units produced in the given shift time.

Sol- WKT-

Cycle Time, $T_C = \frac{T_n}{P_w}$

WKT. $T_n = 3.48$ mins, $P_w = 125\% = 1.25$

$\therefore T_C = \frac{3.48}{1.25} = 2.78$ mins

$\therefore$ Cycle Time, $T_C = 2.78$ mins.

WKT-

Daily Production Rate (R_p) = $\frac{\text{Actual Time Worked}}{T_c}$

$\therefore$ Actual Time Worked = 6.95 hrs

$\therefore R_p = \frac{6.95 \times 60}{2.78}$

$\therefore R_p = 150 \text{ units} = \varphi$

(e) Continuing, determine Worker's Efficiency & Standard Hours if the worker's performance is at 125%.

Sol-

Here -

$H_{std} = \varphi \cdot T_{std}$

WKT- $\varphi = 150$ units

$\therefore H_{std} = 150 \times 4$

$\therefore H_{std} = 600 \text{ mins} = 10 \text{ hours}$

& WKT-

$E_w = \frac{H_{std}}{H_{sh}}$

Here $H_{sh} = 8$ hrs

$\therefore E_w = \frac{10}{8} = 1.25 = 125\%$

④ Continuing, Suppose the worker's pace is at 125% but now the actual time worked is 7.42 hours. Determine the no. of pieces produced, no. of Standard Hours & worker's Efficiency.

Sol-

New Actual Time = 7.42 hours

$P_W = 125\% = 1.25$

WKT-

① No. of Units (Q) = $\frac{\text{Actual Time Worked}}{T_C}$

Here $T_C = 2.48$ mins

$\therefore Q = \frac{7.42 \times 60}{2.78} = 160$ units

② WKT-

$H_{Std} = Q \cdot T_{Std}$

$H_{Std} = 160 \times 4 = 640$ mins $= 10.67$ Hours

③ WKT-

$E_W = \frac{H_{Std}}{H_{Sh}}$

Here Shift = 8 hours

$\therefore E_W = \frac{10.67}{8} = 1.33 = 133\%$

∴ Worker's Efficiency has increased by 33%

② Hybrid Systems -

These are work systems that have a worker element as well as a machine element to achieve a task/job.

Q. What are the various classifications of Hybrid Systems?

Ans - Hybrid systems are classified as -

① Types of Powered Machinery

Here it is again classified as -

ⓐ Portable Power Tools - Eg: Drilling Machine

ⓑ Mobile Power Tools - Eg: Vaccum Cleaners

ⓒ Stationary Power Tools. - Eg: Lathes.

↳ ⓐ Portable Power Tools -

- They are light weight & can be moved by hand from 1 location to the other.

↳ ⓑ Mobile Power Tools -

- They again classified as -

(i) Transportation Equipment
↳ Eg: Taxis, Planes

(ii) Transportable & Mobile During Operation -
↳ Eg: Tractor

(iii) Transportable & Stationary During Operation -
↳ Eg: Generator.

↳ (c) Stationary Power Tools -

- They are classified as -

(i) Machine Tools - Eg: Drilling & Milling M/c's.

(ii) Office Equipment - Eg: Printers, Fan etc.

(iii) Other Machinary - Eg: Sew M/c's, Ovens, Furnaces

(2) No. of Workers vs No. of Machines -
Here it is again classified as -

(i) One Worker - One Machine

(ii) One Worker - Many Machines

(iii) Many Workers - One Machine

(iv) Many Workers - Many Machines.

(3) Level of operator's Attention Required -
Here it is classified as -

(i) Full Time Attention

(ii) Part Time Attention during Each Work Cycle

(iii) Periodic attention with regular servicing

(iv) Periodic attention with random servicing.

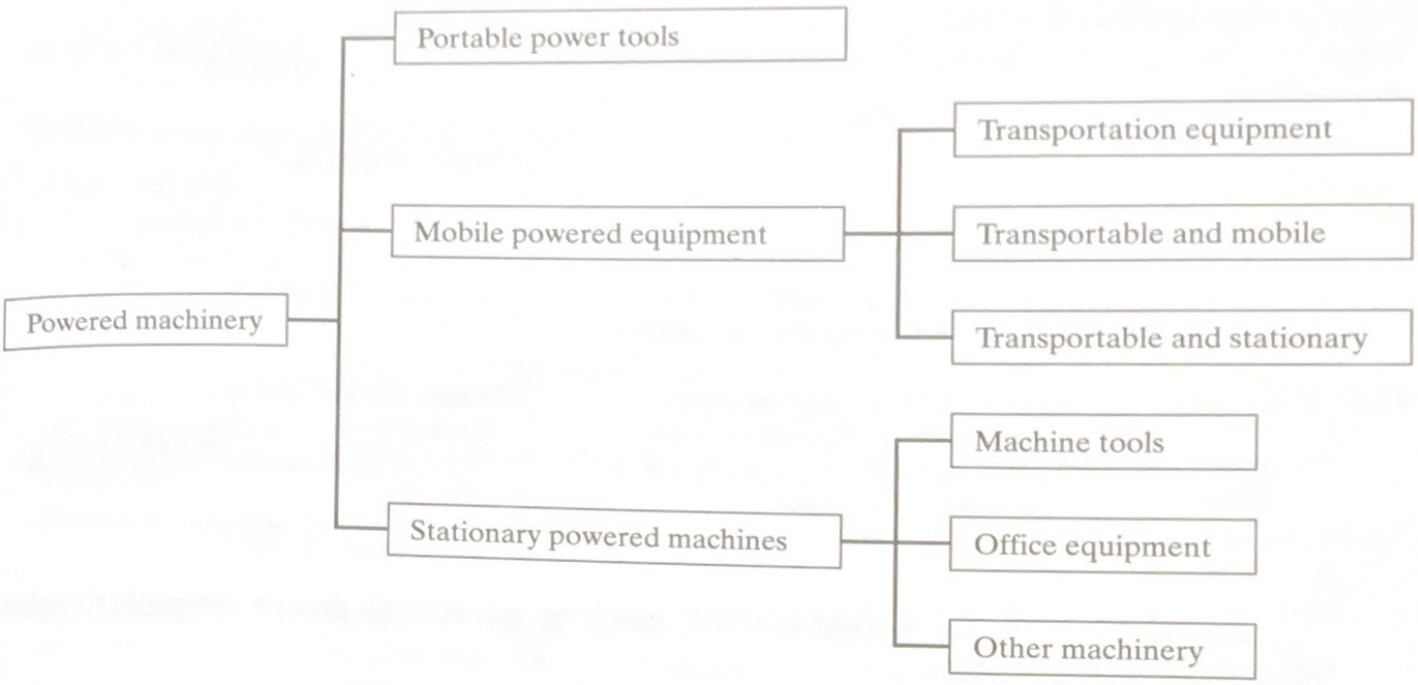

Figure 3 Classification of powered machinery in worker–machine systems.

↳ Cycle Time Analysis for Hybrid Systems -

① Cycle Time when NO overlap is there b/w worker & machine.

(Or)

For Sequential work!

(i) Then for Sequential / No overlapped work, the normal time is -

$$T_n = T_{nw} + T_m$$

Here T_n = Normal Time

T_m = Machine's Cycle Time

T_{nw} = Worker's Normal Time.

(ii) Standard Time in terms of allowance -

Here -

$$T_{std} = T_{nw}(1 + A_{pfd}) + T_m(1 + A_m)$$

Here A_{pfd} = Worker's Allowance

A_m = Machine's Allowance.

Note - Usually & Commonly A_m is set at 30%

② Hybrid Systems ≡ Internal Work Element -
(or)
Internalization:
(or)
For Simultaneous Work!

(i) If the Machine Element ↑ than worker element, then -
Tn & Tstd of the machine are considered

(ii) If the Worker Element ↑ than Machine Element then -
Tn & Tstd of the Worker are considered

Case ① (a) In a Hybrid System, the normal time of Manual Work Element is 1 min. & the semi automated machine time is 2 mins. The pfd allowances are 15% & machine allowance is 30%. Determine the standard time of the work cycle.

Sol: It is a case of sequencially order:

Here $A_m = \frac{30}{100}$ $A_{pfd} = \frac{15}{100}$

$T_{nw} = 1 \text{ min}$ $T_m = 2 \text{ mins}$

WKT- $T_n = T_{nw} + T_m$

$\therefore T_n = 3 \text{ mins}$

WKT- $T_{std} = T_{nw}(1 + A_{pfd}) + T_m(1 + A_m)$

$\therefore T_{std} = 1\left(1 + \frac{15}{100}\right) + 2\left(1 + \frac{30}{100}\right) = 3.75 \text{ mins}$

⑥ Now, if the manual work element & machine work element are being done simultaneously then what is the standard & normal times?

Sol- It is a case of Simultaneous Order-

Here the machine element is greater. Hence-

$T_n = 2 \text{ mins}$

& $T_{m\,std} = T_m (1 + A_m)$

$T_{std} = 2.6 \text{ mins}$

↳ ③ Automated Work Systems -

- Automation - It is technology where is procedure is accomplished without Human assistance
- Automated Work Systems are of 2 types -

① Semiautomated Systems / Machines -

- These machines perform a portion of the work cycle under some forms of program control & a human operator tends to the machine during the remaining time.

Eg- Automated Lathe Machine, which is automatic during operation but depends on a Human for loading & Unloading.

(2) Fully Automated Machines / System -

- These machines can operate for a prolonged time with NO human interactions.

Eg- Such scenarios are seen in Oil Refineries, Nuclear Power Plants etc.

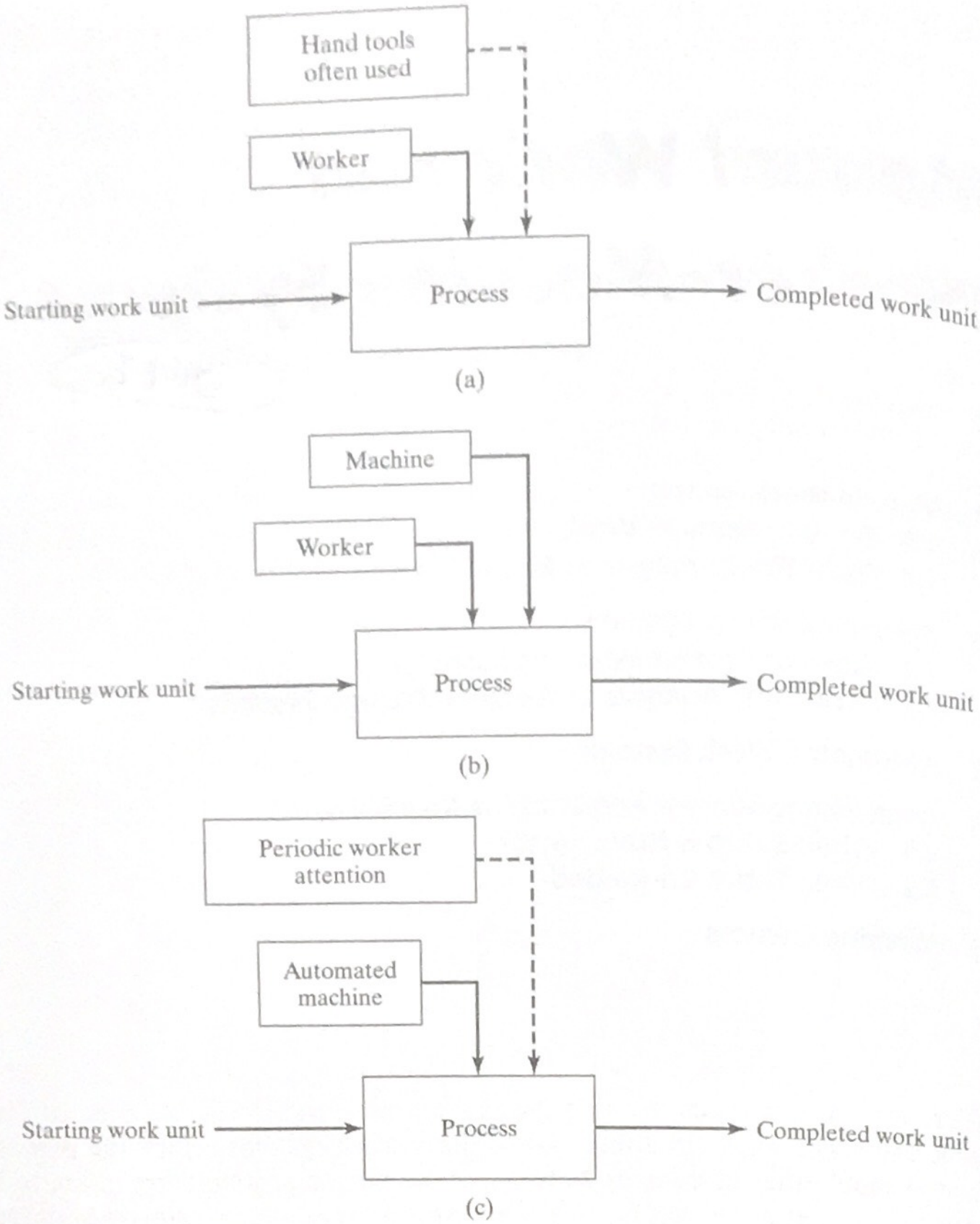

Figure 1 Three types of work systems: (a) manual work system, (b) worker–machine system, and (c) automated system.

⇒ Service Operations -

Q. What are Service Operations?

Ans- Service operations provide a service to a consumer / customer with production operations which is usually a product.

↳ Defferance b/w Service & Product -

Services	Products
① They are INTANGIBLE	① They are TANGIBLE
② Have more Variations	② Have Min. Variations
③ Have a Customa Contact	③ DO NOT have Contact of Customer.
④ Services are Experienced	④ Products are Consumed.

Q. What are the Factors of Service Quality?

Ans- ① Customer Interaction

② Quality of workmanship provided

③ Waiting Time

④ Service Time

⑤ Service Organizations

⑥ Internal Services

⑦ Services from Product Companies.

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Q. ** What are the different types of Services?

Ans- Types of Services include -

① Scheduled Services -

- Here Starting time is known in advance
- But the Service Time is variable.

Eg - College Classes, Office Appointments etc.

② Random Services -

- Here the Customer arrives at any random hours during the available hours for a service

Eg - Hospital Emergency Rooms & Fast Food Outlets.

③ Facilities Based Services -

- Here the Customer must be present at the provider's facility, where the service is rendered.

Eg - Banks, Casinos, Doctor's office, Hotel, Movie Theatres, Restaurants etc.

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(4) Field - Based Services -

- Here delivery of the service is at the doorstep / at the location of the customer.

Eg - Large Household Appln repairs
Waste Management
Telephone services
Plumbers
Electricians etc.

1.2 Service Work

The difference between facilities-based and field-based services is highly correlated with the level of contact between the customer and the service provider. ***Level of customer contact*** refers to the proportion of time that the customer is involved relative to the total time to provide the service. In ***high-contact services***, the customer is involved a high proportion of the service time, and this is generally associated with facilities-based services. Examples include restaurants, hairdressers, dental offices, and prisons. The service worker must use his or her own discretion in dealing with and relating to the customer. Each service is different in this respect, because each customer is different. In ***low-contact services***, there is little direct contact between the customer and the service provider. This is most likely to correlate with field-based services. Examples include postal and parcel delivery, news associations such as Associated Press, and many government services (e.g., fire departments, planning commissions, county offices, tax collectors). Between the two extremes there is a range of ***medium-contact services***, such as police departments, law firms, banks, and automotive repair centers. Some of the important implications of customer contact level on the design of service work and the selection of workers can be summarized as follows [1]:

- In high-contact, random service operations, it is more difficult to plan staffing requirements because the duration of the service time of each customer is not known in advance, and the times and numbers of arrivals are not known in advance. This is not as much of a problem in low-contact service operations, because the service is not performed in the presence of the customer, so the service provider has more control over scheduling.
- With low-contact services, it is possible to analyze the work process, make methods improvements, and achieve efficiencies that are normally associated with factory production operations. In fact, low-contact service operations can often be organized in much the same way as a production process. Such improvements are more difficult with high-contact services due to the close coupling of the service procedure and the customer.
- Different types of workers are required for high-contact services than for low-contact services. For services with high customer contact, workers should have good

TABLE 3 Examples of Facilities-based and Field-based Services

Facilities-based Services	Field-based Services
Bank	Custodial and janitorial services
Restaurant, hotel, casino, movie theater	Temporary manpower services
Retail trade, excluding mail order services	Large household appliance repairs
Hairdresser's shop, barbershop	Waste management (garbage collection services)
Automotive repair shop	Telecommunication (telephone services)
Transportation of passengers (not freight)	Cable TV provider
Doctor's office, dental office	Plumber (repairs)

TABLE 4 Comparison of Manual Work and Service Work

Comparison Factor	Manual Work	Service Work
Processing and flow of work:	Materials	Information, customer experience
Worker attributes:	Physical strength and endurance, manual dexterity, ability to learn a physical task	Mental acuity, communication skills, problem-solving skills, ability to learn a task involving information
Worker moniker:	Blue-collar worker	White-collar worker
Industry categories:	Primary and secondary industries	Tertiary industries
Where the work is performed:	Factories, warehouses, distribution centers, farms, mines, construction sites	Banks, schools, government offices, retail and grocery stores, restaurants, airports, hospitals, hotels, various types of offices

1.3 Standards and Staffing for Service Work

Q. Why staffing is an issue in service operations

Ans –
1. Variable service
2. Random Arrivals
3. Customer contact
4. Intangible work units
5. Undefined work
6. Intermittent service
7. Creative work.

In general, determining time standards for service work is not as str for repetitive production work involving physical labor. Analysis te mining staffing requirements for manual work and worker-machine repetitive work cycles typically found in production work are base of time standards for the tasks performed. The time required to pe ual task can be determined with considerable precision because th attributes of the repetitive work cycle have been standardized.

The following reasons explain why setting standards and deter els are more difficult for service operations:

- *Variable service.* Services of a given type vary, and therefore the time to provide the service varies. Even services that are provided on a repetitive basis are variable. For example, the time for a car salesman to sell a car to a prospective buyer is highly variable. In some cases, several hours may be invested to sell the car. In other cases, the car is sold in much less than an hour. Oftentimes, the salesman invests several hours, but the outcome of the effort is no sale.
- *Random arrivals.* In many service operations, customers arrive at random intervals, making it difficult to predict workloads and determine appropriate staffing levels.
- *Customer contact.* The level of customer contact affects the service time. In general, services characterized by a high level of customer contact tend to be more variable, because the customer's presence affects the time to complete the service.
- *Intangible work units.* The work units in service work are intangible and sometimes difficult to define, and therefore the problem of defining a standard method and standard time is difficult.
- *Undefined service.* Service work may consist of a variety of tasks whose exact details are not known in advance, and therefore a standard method cannot be specified. An example is nonroutine automotive repair work, such as work requiring diagnostics analysis (e.g., fixing a rattle in the car door). Service work of this kind is often charged simply by the time it takes a worker to complete a job, usually at an hourly rate.
- *Intermittent services.* Some services are delivered on an ongoing and intermittent basis. There is a continuing relationship between the service provider and the customer, but the interactions do not occur continuously. An example is an investor's account executive at a stockbroker. The investor and the account executive have a continuing relationship that requires only occasional contact, usually by telephone, to buy or sell securities. Standards in this kind of work are impractical.
- *Creative work.* In service work that requires creativity, the work cannot be directly measured. Therefore, time standards for creative work are not practical. The results of the work are usually measurable, but not the work itself. Evaluation of knowledge workers is therefore best assessed by the quality of what they produce, not by the methods of working.

⇒ Office Work -

- Office - It is a place at which the business-oriented activity or services of a company are rendered.
- Office Work - It is the work concerned with the business functions such as design, sales, accounting, Scheduling & administration.
- Offices are also Entities with which both Consumers & manufacturing interact.
- Offices can also consists of Call Centres & BPO's. (Business Process Outsourcing).
- Onshore offices - They are offices which are in the SAME country as the customer.
- Offshore offices - They are offices which are in DIFFERENT countries from that of the Customer.

↳ Office Activities Vs Office Applications

① **Office Activities** - It is the physical & mental actions performed by a office worker while carrying out an assigned task.

eg - Typing, Filing Paper etc., Calculations. Photocopying Proofreading, Steteling etc.

② **Office Applications** - It is related Business functions & it is oriented to the End Results or achievements.

eg - Photocopying, Invoices, Payrolls, Presentations, Sales forecast etc.

↳ Creative Work Vs Routine Work.

① **Creative work** -
- It is Intellectually more difficult.
- Requires special skills
- Involves learning, analysis, decision making & problem solving.

② **Routine Work** -
- It is REPETATIVE & Less difficult
- Lesser Educational Requirements.

TABLE 6 Examples of Office Activities

Analysis (e.g., analysis of data)	Making telephone calls
Answering telephone calls	Participating in meetings
Calculating	Photocopying
Collating and sorting	Proofreading or checking (e.g., a document)
Decision-making	Reading
Drawing or sketching	Thinking
Drawing conclusions or inferences	Typing or keying (e.g., personal computer)
Filing papers	Writing (with pen or pencil)
Mail handling	

Source: Based on lists in [2] and [3].

TABLE 7 Examples of Office Applications

Accounts payable	Presentations
Accounts receivable	Price lists
Bills or material (parts list for products)	Production scheduling
Engineering drawing preparation	Proposals (e.g., proposal to a customer)
Inventory lists	Purchase orders
Invoices	Reports (e.g., research report)
Payroll	Sales forecasts

Source: [3].

↳ Knowledge Workers Vs Support Workers.

(1) Knowledge Workers - • They accomplish "Creative Information Processing Activities".

• They are White Collar professionals.

eg - Managers
Lawyers
Engineers
Designers
Researcher etc.

- Knowledge workers are again classified into -

(i) Managerial (ii) Non - Managerial.

(2) Support Workers - • They provides administration & staff assistance.
• Serve the needs of knowledge worker staff.

eg - Secretary, Peon, Stenographer, Typist, Clerk, Receptionists etc.

TABLE 9 Typical Titles of Knowledge Workers Who Work in Offices

Managerial Titles		NonManagerial Titles	
Chief executive officer	Principal (school)	Engineer	Research scientist
President	Provost (university)	Senior engineer	Professor
Vice president	Dean (college)	Lawyer	Editor
Manager	Department chairperson	Doctor	Chemist
Superintendent (schools)	(academic department)	Dentist	Sales/marketing analyst

	Routine work	Creative work
Knowledge workers	0%	100%
Support staff	100%	0%

Figure 1 Ideal allocation of workloads between knowledge workers and support staff.

	Routine work	Creative work
Knowledge workers	63%	37%
Support staff	93%	7%

Figure 2 Allocation of workloads between knowledge workers and support staff as indicated in a 1988 study reported in [3].

↳ Comparision of Knowledge – Information – Data :-

① **Data** - It is raw facts & statistical data, with an interpreable meaning.

② **Information** - It is the data that is acquired with revalance & meaning.

③ **Knowledge** – It is the Information which is learned & synthesized with other information in the Human Brain.

↳ Office Automation vs Office Augmentation -

① Office Automation - It is term used to describe the use of computers & technology to automate repetitive & routine work in an office environment.

- It involves the Integration of 3 vital office Technologies -

(i) Office Machines - Eg:- Typewriters, Photocopiers

(ii) Data Processing Machines - Eg: Computers, Printers

(iii) Communication Machines - Eg - Telephones, Fax etc.

② Office Augmentations - It is said to aid the creative work aspect of a worker with the help of technological devices.

Eg - Computers aid to Design etc.

↳ Comparison b/w Office Automation & Office Augmentation

Automation	Augmentation
• Routine Work	• For Creative Work
• Used by Support workers	• Used by Knowledge Workers
• It has ↑ efficiency	• It has ↑ effectiveness.

Computer application:	Automation	Augmentation
Type of work:	Routine work	Creative work
Type of office worker:	Support personnel	Knowledge workers
Attributes:	High efficiency Less time consuming Less costly	High effectiveness Convenience to worker High quality results
Office application examples:	High-volume tasks Database applications	Designing products Creating new text

Figure 3 Automation and augmentation for routine and creative work.

UNIT-2

↳ Chapters from TB -

(8), (9), (10).

8 → Intro to Methods Engg.

9 → Charting & Diagramming Techniques

10 → Motion Study & Work Design.

Introduction to Method's Engg & Operational Analysis.

↳ Methods Engineering –

- It is also –
  - Work Study
  - Work Simplification
  - Methods Study
  - Process Re-engineering
  - Business Process Re-engineering.

Def- It is the analysis & design of work method & systems including the tooling, equipment, technologies & work place layout.

- It aims at –
  1. Increase Productivity & Efficiency
  2. Reduce Cycle Time
  3. Reduce Product Cost.
  4. Improve Customer Satisfactions
  5. Improve Worker's Safety.
  6. Increase Work System Flexibility

Note - ILO (International Labour Orgn) is in Geneva!

↳ Operations Analysis -

by - It is defined as the study of operations or a group of operations for the purpose of analysing their effeciency & effectiveness.

↳ Methods Engineering

- It is a step-by-step process.

↳ Work Measurement -

- It tries to measure the Standard Methods in terms of time.

∴ Methods Engg & Work Measurement have a Complimentariness" & that is to achieve Productivity.

∴ Method Study is the Predesson of Work Measurement.

↳ Terms of Problems being addressed in Methods Engineering :-

In terms of problems arising in Method Engg, it is of 2 types -

(i) Method Analysis

(ii) Method Design.

(i) Methods Analysis - It is the study of an existing method / processes by breaking it down into basic / Elemental pieces of work.

↳ It aims at - / Objectives of Methods Analysis -

(a) Eliminate
(b) Combine
(c) Rearrange
(d) Simplify.

(ii) Methods Design - It is of 2 concerns -

(i) New Methods

- Here there are not precendent analysis
- It is developed for new facilities, equipment & products / service

(ii) Redesigning of Existing Methods.

- Here they might be a precedent analysis
- It is developed to increase productivity & efficiency of the existing methods / processes.

↳ Method's Engineering -

- It always 1st focuses on the Objectives & then on the means to achieve those objectives.
- Method's Engg. helps us to achieve Productivity in Production
- Traditional Focus is to Increase Productivity & Reduce Lead Time
- Now, Contemporary Focus is to Increase & Retain Customer Satisfaction !!

↳ Evolution of Methods Engg -

- It was found by Frank Gilbert & dealt with the motions performed by workers in Bricklaying.
- In the early 20th Century, FW Taylor / Frederick Taylor came up with "The Scientific School of Management".
- George Elton Mayo introduced "Human Relations School of Thought".
- Human Relations School of Thought - Said "Man is a Social Being & not only an economic man".

↳ Agility, Flexibility & Modularity -

(1) Agility in Manufacturing - It is the ability of an organisation to respond quickly & swiftly to any changes in market trends & customer requirements

(2) Modularity in Design - It is the practice where smaller components which are independent of eachother are arranged to produce a whole mechanism.

(3) Flexibility in Manufacturing - It is a system, which allow the system to react in case of changes whether those changes are Predicted or unpredicted.

- Flexibility is only observed with in the working of an Organisation.
- Modularity corresponds to flexibility in Configuration.
- Agility is a outcome of having flexibility in the Work Station.

↳ Factors to consider in Methods Engg are -

① Human Factor - • It constitues of empathy etc.

② Technical Factor - • It concerns to - Technical Investments.
- Skill Sets of operators
- Equipment

③ Economical Factor - • It concerns to choose a "Vital Few" for better economic results as per the '80/20 Rule'.

ie 80/20 Rule - aka - Pareto Principle :

Here 20% of existing methods & 80% of Reworking Methods

↳ Few Statements to note -

(i) "Increase in Productivity does not necessarily mean Increase in Production."

(ii) "Production may lead to Productivity, but it is not a fixed Rule."

↳ How to Apply Methods Engg -

Step ① "Define the Problem & its Objectives"

In this step - ① Problem Visibility always provides an accurate & fitting soln to the problem.

There are 3 factors to be considered -

ⓐ Human factor -

ⓑ Technical factor -

ⓒ Economical factor -

- The problem in Methods Engg. can be identified as -
  - ↳ Low Productivity
  - ↳ High Cost
  - ↳ Ineffecient Methods
  - ↳ New methods req etc.
- Possible Objectives -
  - ↳ Increase Productivity
  - ↳ Reduces Cost & Labour
  - ↳ Improve Safety etc.

↳ Remember -

" Problem clearly defined is halved Solved"

Note - Always mention the Objectives with the problem.

Step (2) "Analyze the Problem":

- It consists of data collection / recording of "as is" & analysis activit
- Here it is to capture the Essence of the Problem -
- Here various tools & techniques are used.

(Read more after New App)

∴ It involves –

1. Identify the basic fn of the operation
2. Gather Background information.
3. Observe Existing Processes

④ Collect Data

⑤ Examine the Data

⑥ Develop Mathematical Models / Simulations

⑦ Then use Charting Techniques.

↳ Step ③ "FORMULATE ALTERNATIVES".

- There are many ways which maybe more efficient to achieve a Task.
- Here we must formulate / record all the possible alternatives
- It is NOT focused on the Best Alternative

↳ Step ④ "Evaluate Alternatives & Select the Best".

- It involves a Methodological assessment of all the possible alternatives from which the best one is selected.
- Here a systematic procedure is used to find the BEST alternative.
- Each alternate has its own strengths (Pros) & weaknesses (Cons).

- For each alternative - There are 2 types of features & specifications.

  (a) Must Feature - These are crucial / pivotal features that the alternatives must have.

  (b) Desirable Feature - These specification are NOT crucial to the problem but are rather desirable.

- finally a "CRITERIA MATRIX" is used to find the Best Alternative.

- All the alternatives must satisfy "Must Features" otherwise they are dismissed.

- Then all the alternatives are compared based on their "Desirable features".

- A "Max. Possible Point Score" is allocated to each "Desirable feature".

- Without evaluation, selection of the "lowest cost bidder" is a MISTAKE!

↳ Step (5) "Implement the Best Method".

- Here the proposed changes must be installed in the existing / new processes.

- It may be subjected to Pilot Trials & simulations

- Implementation includes " COMPLETE DOCUMENTATION of the proposed solution.

↳ Step ⑥ " Audit the Study ".

- It is desirable to perform a follow up on whether if the original problems and objectives are resolved with the new process.

∴ Tools are broadly classified as - (a) Specialized Tools

(b) Generic Problem Solving Tools.

① SPECIALIZED TOOLS

↳ Tools & Techniques of Method's Engg used to Analyze a Problem -

(i) Charting & Diagraming Techniques

(ii) Motion Study & Work Design

(iii) Facility Layout Planning

(iv) Work Measurement Techniques

(v) New Approaches in Method's Engineering.

} Specialized Tools.

1 **Charting and Diagramming Techniques.** There are many charting techniques available for collecting, displaying, and analyzing data on a given work system or operation sequence of interest. They can be classified into the following categories:

- *Network diagrams.* These are used for analyzing work flow, assembly line balancing, and project scheduling. Special algorithms are often available to analyze these network diagrams.
- *Traditional industrial engineering charting techniques.* These are used to symbolize and summarize the details of an existing operation or sequence of operations. The traditional charting techniques can be used to analyze the activities of one human worker, groups of workers, worker-machine systems, materials, parts, and products.
- *Block diagrams and process maps.* These diagrams represent alternative ways of depicting processes. They are sometimes used in place of the traditional IE charting techniques.

(ii) MOTION STUDY & WORK DESIGN -

- It focus on "Basic Motion Element" of a worker.
- Basic Motion include -

(a) Reach

(b) Grasp

(c) move

(d) release

- There are 17 Basic Motion Elements which are mostly achieved by Hands.
- Here unnecessary motion can be eliminated
- It highly co-relates to "Principles of Motion Economy".

(iii) FACILITY LAYOUT PLANNING -

- It deals with the utilization of infrastructure.
- A facility can be a "House, building, factory, workshop" which a company either owns or rents it out.
- Facility Layout - Refers to the size & shape of a facility.
- It behaves as a "Blueprint for a given location / facility".
- Most employeed planning is "Systematic Layout Planning" introduced by Richard Muther.
- Tools used are -
  - (a) String Diagrams - It is complemented with FPC's.
  - (b) Flow Diagrams - They show the distance & activities at each location.
  - (c) Travel Diagram
- Layout plays a significant role in determining the efficiency of operations achieved.

(iv) WORK MEASUREMENT -

- It is aka "TIME STUDY".
- Work Measurement is complimented with Methods Engg.

- Commonly used Tools are -

  (a) PMTS (Predetermined Motion Time System)

  (b) MTM (Methods Time Measurement)

  (c) MOST (Maynard operation Sequence Technique)

  (d) Work Sampling -

↳ (a) PMTS -

- It is a database of basic motion element & their corresponding normal time.
- It's principle appⁿ is to determine "Standard Time".

↳ (d) Work Sampling -

- It is a statistical technique for determining proportions of time spent by a worker / machine in various categories of activity.
- It can also be used to find - (1) Machine Utilization

  (2) Worker Utilization

(V) New APPROACHES IN METHODS ENGG-

- It includes new philosophies such as -

(a) Lean Production System (LPS) - which was introduced by Toyota Production Systems.
- It focuses on Elimination of wastage in a production setup.

Eg - Value stream Mapping.

(b) Total Quality Management (TQM) - Here it involves a Customer from start to finish.
- It utilizes the TOP → DOWN approach.

(c) Six Sigma - • It is a Product / Service Level Quality aspect.
- It measures the defects in a Population.
- It allows only 3.4 defects in 1 million samples
- It was introduced by Motorola & is an Outcome.

(2) GENERIC TOOLS

↳ Generic Problem Solving Tools –

- They are the tools that can be used in any Engg.
- It is to exhibit data & so that it can be readily interpreted.
- These tools play a vital role in "Statistical Process Controls (SPC)".

(1) Histograms: (aka Bar Graph)

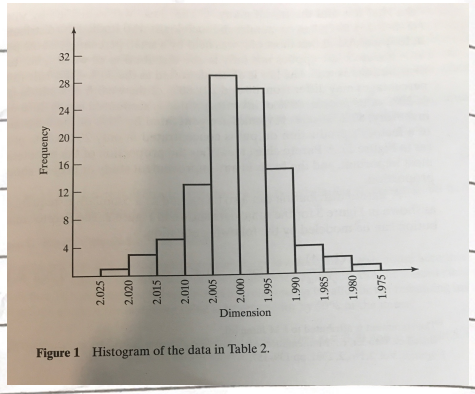


Figure 1 Histogram of the data in Table 2.

- It is a bar graph.
- It charts "The frequency in proportions of the length of the Bar".
- It is a Graphical display of "Frequency Distribution".

↳ Formula of Pareto Chart -

① $y = \frac{(1+A)x}{A+x}$ for $0 \le y \le 1$ & $0 \le x \le 1$

② $A = \frac{x(1-y)}{y-x}$

If $A = 0$, $y = 1 \ \forall x$

If $A = \infty$, $y = x$

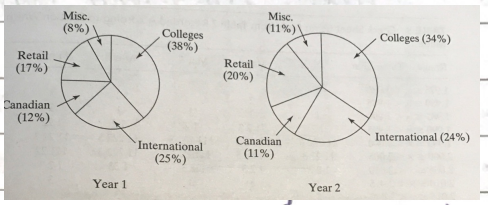
Here y = Cumulative Fraction of Value Variable

x = Cumulative Fraction of Item Variable

A = Constant.

(2) Pareto Charts :

- It is a special form of Histogram
- It focus on "Vital Few & Trivial Many".
- It is also known as => (80/20) Rule, Pareto Rule. (y/x)
- It helps to create "Prioritisation".
- It gives the Graphical Display to the "Tendencies of a vital few to be more valuable than the Trivial Many".



(3) Pie Chart :

- The relative areas of segments are proportional to the mag^{ns} of data that comprise the whole circle.
- Multiple Pie Charts can be displayed.

(4) Checksheet :

- It is a Data Gathering Tool & generally used in preliminary stages of a study.
- Here Tally Marks are used to arrive at a definite frequency.

(5) Defects Concentration Diagram (DCD)

- It is used to analyse a cause of defect in a product
- It conveys the location of the defect.

(6) Scatter Diagram (aka XY Plots)

- It is used to identify any POSSIBLE relation b/w 2 processes variables.
- Here for each value of x there is a corresponding value of y.
- Caution to be maintained in Scatter plot is that there may be more variables with larger importance than the variables plotted.
- The Independent variable is (x) & dependent variable is (y).
- It involves the predictions of CO-RELATIONS -
  - (a) -ve Co-Relation (Inversly Proportional)
  - (b) +ve Co-Relation (Directly Proportional)

classmate
Date
Page

(7) Cause Effect Diagram (aka Fishbone Diagram)

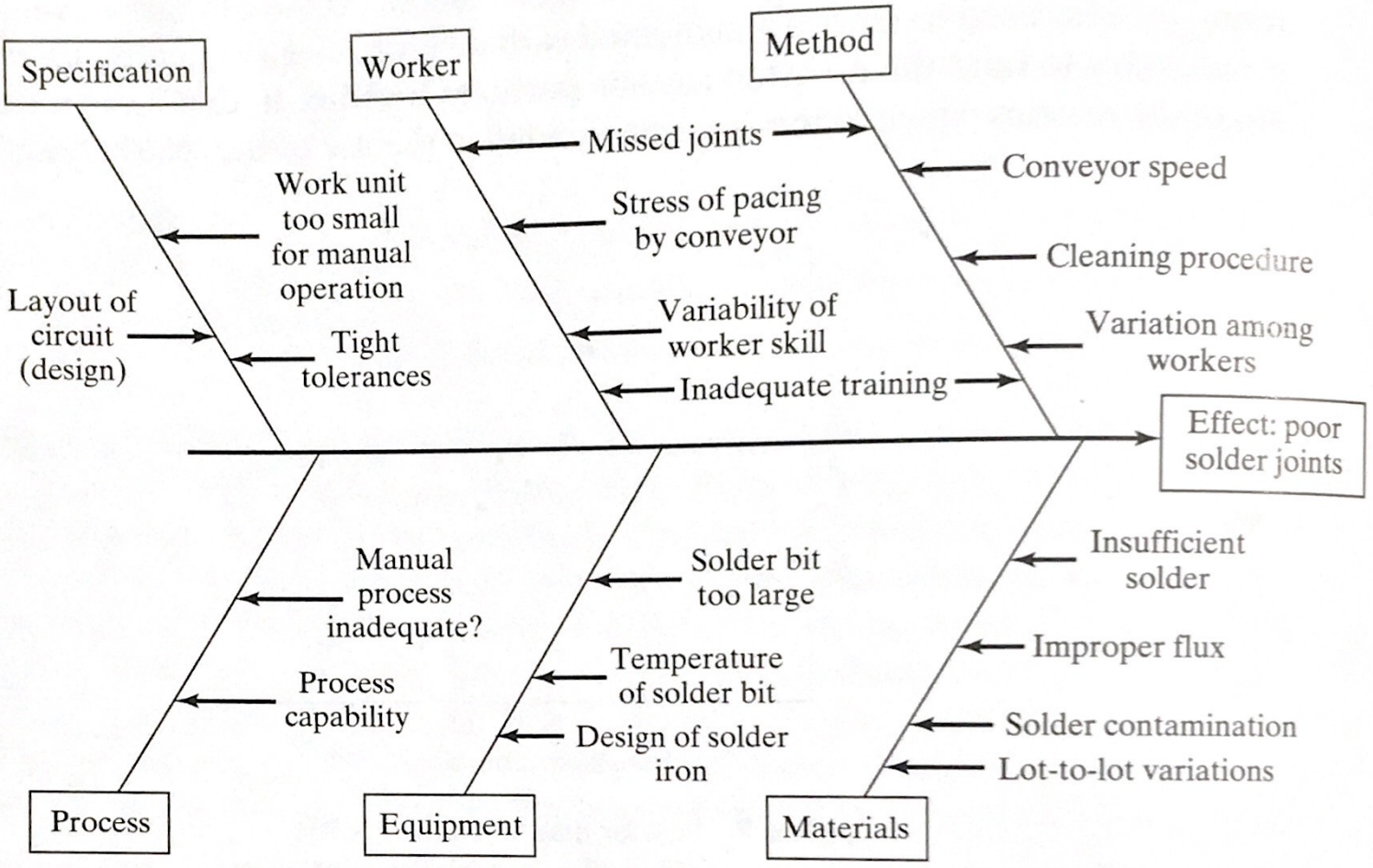
(aka Ishikawa Diagram)

- It is a Graphical Tabular chart used to list potential causes of a problem.
- Developed by Work Teams who study operational problems.
- The 6 general causes are -

~~1. Man~~
~~2. Machine~~
~~3. Material~~
~~4. Money~~

1. Machines : Equipment
2. Materials : Starting Materials
3. Methods : Sequence of Activities
4. Mother Nature : Env. Factors.
5. Measurement : Validity & Accuracy
6. People : Human Factor.

Eg of Fishbone Diagram.



How to Automate in Method's Engg.

- There are 3 approaches / principles used in Automation in Method's Engg.

  1. USA principal
  2. 10 strategies of Automation
  3. Automation Migration strategies.

↳ ① USA Principle –

U = Understand
S = Simplify
A = Automate.

- Some kind of charting tools are used to identify operations that can be automated.
- Mathematical models can be useful to indicate the relationship b/w I/P parameters & O/P variables.
- Simplification involves a checklist.
- The process in its simplest form "can be automated" based on the 10 strategies of Automation

↳ ② 10 strategies of Automation – (SCS III OPPC)

- The following are used to search for improvements.

① Specialization of Operations – Involves use of M/c designed to perform 1 operation.

② Combined Operation – Here more than 1 operation is performed at a workstation

③ Simultaneous Operations – Here operations which are combined can be performed sequentially.

④ Integration of Operations – Involves linking several work stations in a single mechanism

(5) Increased Flexibility - It tries to achieve max. utilization of Man & Machine which can be automated by Computer Systems.

(6) Improved Material Handeling & Storage - Here by automation large amount of non-productive work can be eliminated. Thus leading to shorter lead times.

(7) On-Line Inception - It tries to achieve "Corrections during the processes, to avoid any rejection which would lead to wastages".

(8) Process Control & Optimization - Here Individual Process Times can be reduced & quality can be improved.

(9) Plant operation Controls - It is associated a control at a "Plant Level"
- It involves a high level of computer networking in the facility.

(10) Computer Integrated Manufacturing (CIM) - It involves extensive use of computer apps, databases & networking in the entire facility.

The strategies must not be considered Mutually Exclusive (Ex)

↳ ③ Automation Migration Strategies -

- Adopting the Agility of Manufacturing, if a product is successful, leading to a large demand.
- Then any manual operations involved must be tried to do via automation.
- These improvements / automatisations are carried out in many phases.
- There are -

(a) Phase (1): Manual Production:

This is used for new products as it is low cost & quick to get started with.

(b) Phase (2): Automated Production:

- Can be done using single cell automated systems working independently
- Here work units are still moved b/w workstations manually.

(c) Phase (3): Automated Integration Production:

- Here multiple stations are automated & movement of work units are also automated.

Q. What are the advantages of such strategies? (Automation migration strategies)

Sol- (1) It allows introduction of new products in the shortest possible time

(2) It allow automation to be introduced gradually.

(3) It avoids commitment to a high level of automation, incase there is a drop in demand of a product.

Charting & Diagramming Techniques.

(i) Charting & Diagramming –

They are classified as - (a) Network Diagram

(b) Traditional Industrial Engg Charting Techniques

(c) Block Diagrams

↳ Objectives of Charts & Diagrams -

1. For work processes to be communicated more readily.

2. Allow the use of algorithms specifically designed for a particular diagramming technique.

3. To divide a work process into constituents elements.

4. Provide a structure in the search for improvements

5. To represent a posed method.

Q. How does an Analyst create a chart?

Ans - 1. Analyst is familiar ⊂ the process.

2. Analyst observes & records info - about the process

3. One - on - One Interviews of those who are familiar with the process

4. Group Interviews of those familiar with the process.

Q. How are Charts analysed ?

Ans:

1. Algorithmic :
2. Checklists :
3. Brain Storming :
4. Separating Value - Added from Non - Value Added operations

Here Max - No. of Arrows possible = $n(n-1)$

where n = no. of nodes.

(a) Network Diagrams -

- They are used for -
  - Project Scheduling
  - Work Flow Analysis
  - Assembly Line Balance.
- They are drawn with ~~Nodes~~ Nodes & Arrows.
- They are used to model the order of Precedance.
- CPM = Critical Path Method
- PERT = Program Evaluation & Review Technique

(b) Traditional IE charting techniques -

- Here 5 symbols are used -
- "Materials, Equipment & Human Operator are the fundamental Elements of a Process"
- An Act done on the material is known as Operation
- Based on Organization these charting techniques are classified as -

(I) Outline / Operations Process Charts (OPC)

(II) Flow Process Charts (FPC)

(III) Flow Diagrams

(IV) Activity Charts.

(I) OPC (operational Process Charts)

(I) OPC – • It is the "Bird's Eye View" for a process.

• It only considers "Operations & Inspections".

• It is a graphical & symbolic representation of operations of a process.

• It consists of series of vertical stems, & hence shows the sequence of operations.

• It has inspections at regular internals.

• It uses only 2 symbols - ie ○ → Operations

□ → Inspection

• Right most column is h/s "Chassis of the assembly".

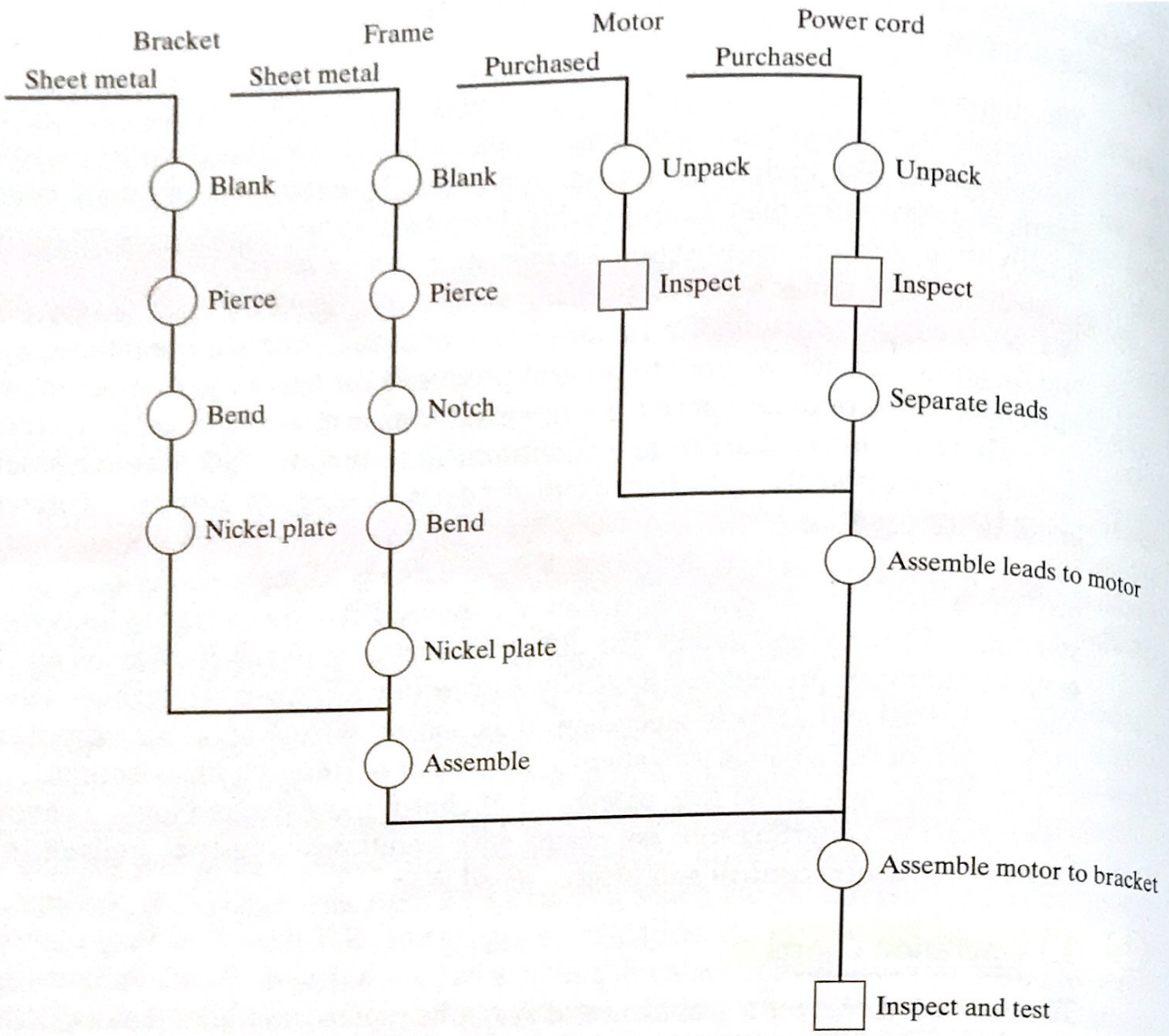


Figure 1 An operation chart for the subassembly of a product.

TABLE 1 Symbols Used in Operation Charts

Symbol	Letter	Description
○	O	***Processing or assembly operation.*** Processing operations consist of changing the shape, properties, or surface of a material or workpart. Assembly operations join two or more parts to form an assembly.
□	I	***Inspection operation.*** An inspector checks the material, workpart, or assembly for quality or quantity.

- Types of Process Charts -

1. FPC
2. worker Process Chart
3. Form Process Chart

3.2 Process Charts

A ***process chart*** is a graphical and symbolic representation of the processing activities performed on something or by somebody. The chart consists of a vertical list of the steps performed on or by the work entity using various symbols to represent operations, inspections, moves, delays, and other activities. The principal types of process chart are: (1) the ***flow process chart***, used to analyze a material or workpiece being processed, (2) the ***worker process chart***, used to analyze a worker performing a process, and (3) the ***form process chart***, used to analyze the processing of paperwork forms. These charts are described in the following sections.

(2) FPC (Flow Process Charts)

(II) FPC -

- It represents the processing activities carried out by "Something or Someone".
- Symbols used are -

(1) Operations → O → ○

(2) Inception → I → □

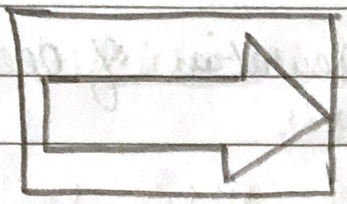
(3) Transportation → M → ⟹

(4) Delay → D → D

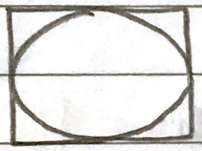
(5) Storage → S → ▽

- symbols can be combined if 2 or more processes happen simultaneously

Eg -



= Inspection & Transportation



= Inspection & operation etc.

- This technique is also used only for a single material or work part.

- In Flow Process Charts, the symbols of FPC are unable to capture any office purposes.

- Flow Process Charts of 3 types -

  (a) Man Type - It records the activities carried out by a human.
  eg - Working, Operating, Waiting etc.

  (b) Material Type - It records activities on a material
  eg - Loading, unloading etc.

  (c) Equipment Type - It records the activities performed by a machine.

(2)

Worker Process Chart. The ***worker process chart*** is used to analyze the activities of a human worker as he or she performs a task that requires movement around a facility.[4] The symbols are virtually the same as those appearing in Table 3, but they are interpreted in terms of what the human worker does rather than what is done to a material. Table 5 summarizes the interpretations. The storage activity is difficult to interpret in the context of human work activity, so it is omitted. Analysis of the worker process chart in the search for improvements involves the same kinds of questions as in the flow process chart, only in the context of the worker performing the task of interest.

In Worker process Chart only 4 symbols are used.

Storage (▽) is removed here.

TABLE 5 Symbols Used in the Worker Process Chart

Symbol	Letter	Description
○	O	***Operation*** performed by a worker at a single location or workstation. The operation may involve movements of materials within the workstation.
□	I	***Inspection***, either to check for quality or quantity, performed by a worker at a single location or workstation.
→	M	***Move*** in which the worker moves from one location to another as a regular element required in the task. It does not include moves within a workstation.
D	D	***Delay*** of the worker. Worker is forced by the situation to wait (e.g., waiting for an elevator). The waiting may involve moving, but the move is not a regular element required in the task (e.g., worker goes to the coffee machine while waiting for the elevator).

3

(III) FORM PROCESS CHARTS -

- They are used to analyze work forms & office procedures
- Symbols for FORM PROCESS CHARTS are -

1. Operation - O → ○ : Punches cards, data entries, filling forms, photocopying
2. Creation - C → ⊙ : Origin of a form
3. Inspection - I → □ : Read information
4. Move - M → ⇨ : Movement by Man or Human Carrier
5. Storage - S → ▽ : Filing, long waiting
6. Delay - D → D : Waiting in a basket
7. Dispose/Destroy - X → X : Form is destroyed.

↳ Flow Diagrams -

- It is a drawing of the facility layout with addition of line representing movement of materials / workers in a facility.
- Arrows are used to indicate movement.
- Flow Diagram are used also along with "PROCESS CHARTS".
- Few Flow Problems are -
  1. Excessive Travel
  2. Backtracking
  3. Inefficient Layout
  4. Traffic Congestion.

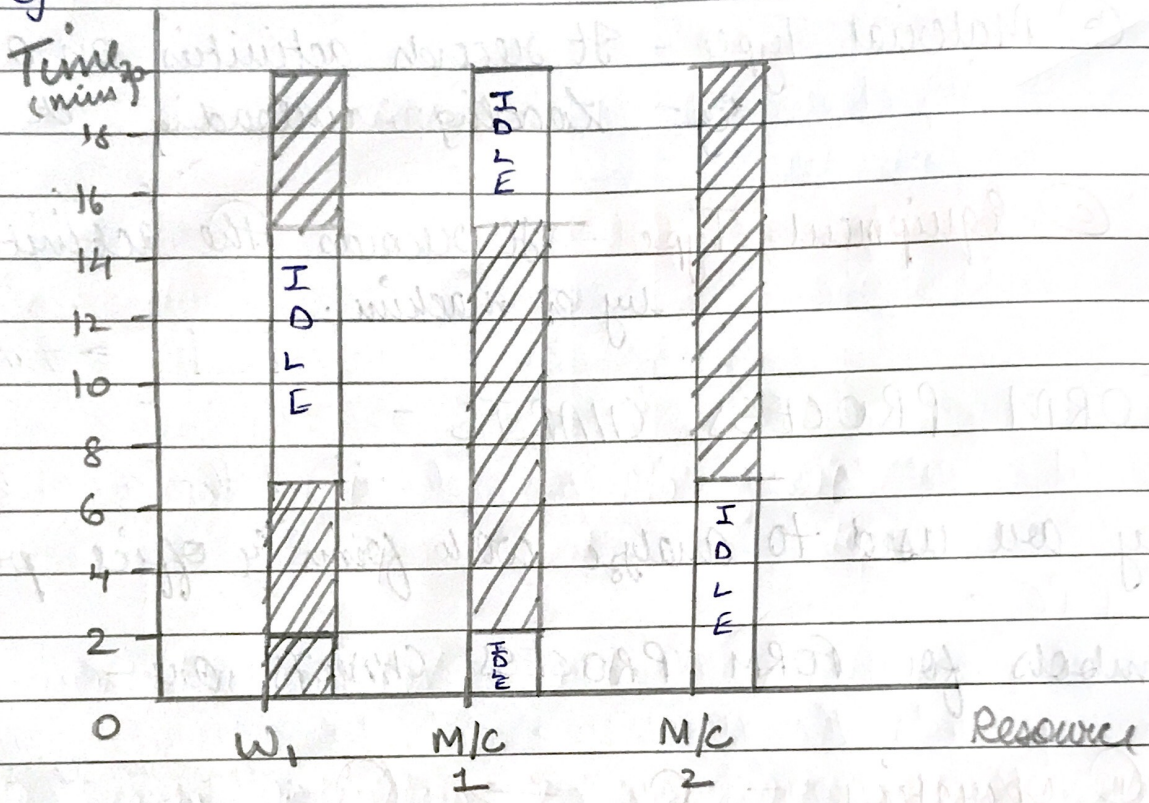
(IV) Multiple Activity / Activity Charts –

The different types of Activity Charts are –

(a) Worker – Multiple machine Charts –
(aka Machine Cluster)

- Here there is only 1 worker but many machines

Eg –



- To calculate the percentage of Utilization of each component –

- M_1 / W_1 ⇒ worked for 12 mins

∴ % of Utilization = $\frac{\text{Time Worked}}{\text{Total Time}} \times 100$

∴ % of Utz^n of W_1 = $\frac{12}{20} \times 100 = 60\%$

$\therefore$ % of Utz^n of M/C$_1$ $= \frac{13}{20} \times 100 = 65\%$

$\therefore$ % of Utz^n of M/C$_2$ $= \frac{13}{20} \times 100 = 65\%$

- Here a work cycle must be developed to minimize "MACHINE INTERFERANC".

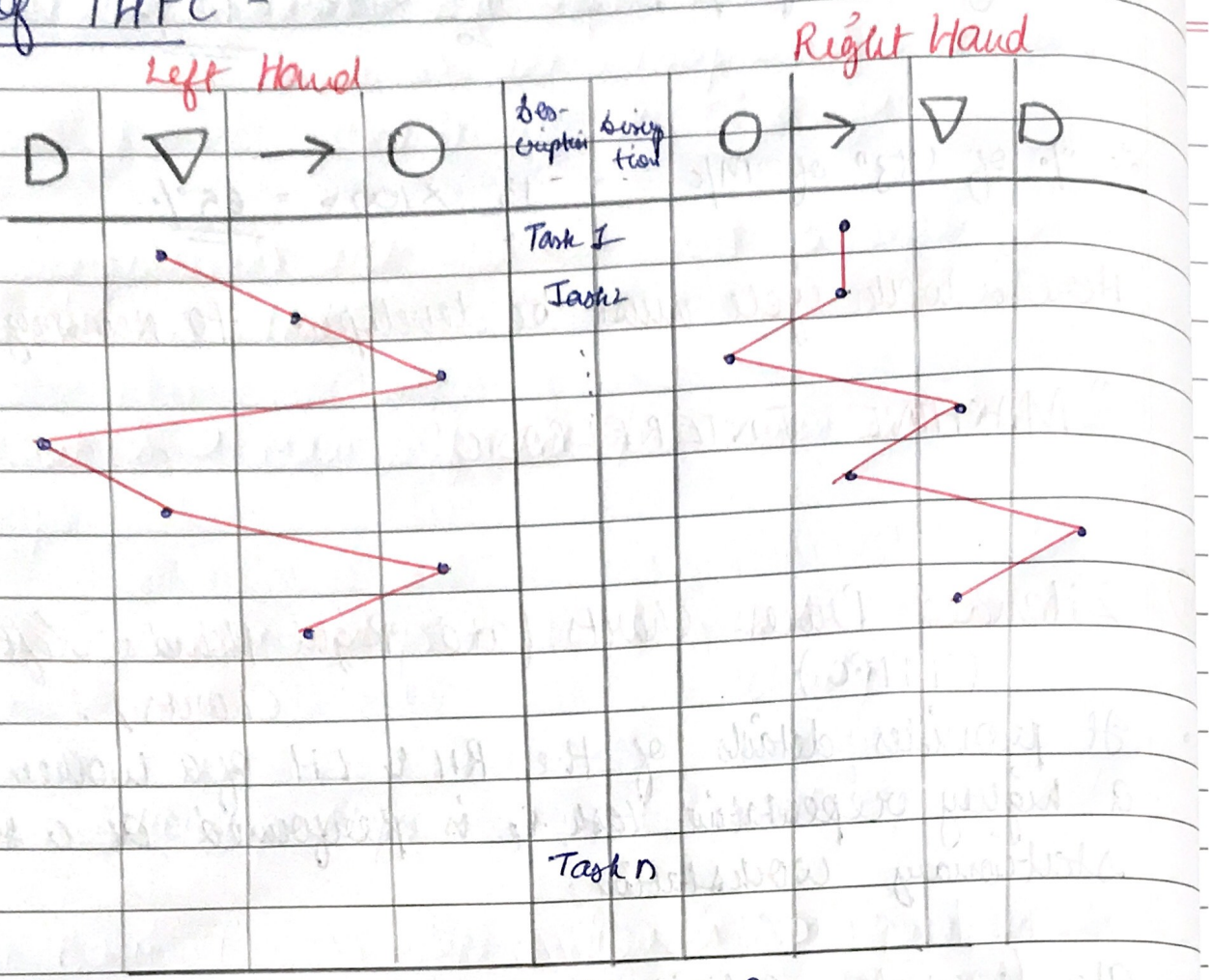
Shading	Color	Activity
Black	Blue	***Operation***: Performing an operation. Worker operating on or handling material at workplace. Machine performing an operation on automatic or mechanized cycle.
Gray	Yellow	***Inspection***: Worker performing an inspection, to check for either quantity or quality.
White (blank)	White (blank)	***Idle time***: Worker or machine is idle, waiting, or stopped.
Diagonal lines	Green	***Moving***: Worker walking outside immediate workplace (e.g., to fetch tools or materials).
Horizontal lines	Red	***Holding***: Worker holding an object in fixed position without performing any work on it.

Figure 6 Shading formats for activity charts.

(b) 2 Handed Process Charts (aka Right Hand - Left Hand Charts) (THPC).

- It provides details of the RH & LH of a worker performing a highly repeatative task & is performed at a single & stationary workstation.
- aka (workplace activity chart)
- Here the Element of Inspection is NOT there.
- It provides information to "Principles of Motion Economy with reperance to the Human Body".
- It revolves around "comfortability in a work Envelop"
- Here the "as is" is recorded by determining the load in both the hands.

↳ Eg of THPC -



No. of LH Step = No. of RH Steps.

(c) Gang / Multiworker Charts -

- It tracks activities of 2 or more workers working in a team.
- It is used to bring better co-ordination & balance work load among workers.

(d) Worker - Machine Chart -

- It displays the work elements & associated times for repetitive systems for a worker & machine setup.

- It is used to reduce sequential activities into simultaneous activities.
- It aims to replace External Work Elements by Internal Work Elements.

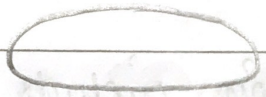
Note -

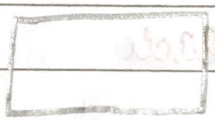
① Activity charts have more than 1 time scale.

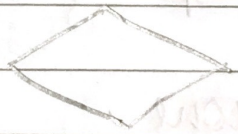
(c) Block Diagrams - • They are commonly applied in Linear Theory to show a relation

- They are commonly applied to represent sequential operations.

↳ Swimlane Charts - (Form of Process Maps)

- Process Maps are widely applied to business processes.
- It is also applicable to production, logistics & service ops.
- Process - It is defined a sequence of Tasks / activities that add value to 1 or more I/p's to produce output.
- Symbols used - (1)  → Stating / Ending of a Process

(2)  → Task / Activity Step

(3)  → Decision Point.

- Swimlane Charts aka Cross Functional Process Charts.

- Swimlane Charts – It depicts how the sets of a process are accomplished by various departments or other functional groups.

- Departments are list in rows & separated by "DASHED LINES."

UUUUVIMP

Q. What is a Main Differance b/w Swimlane & Relationship maps?

Ans – In Swimlane Maps – ▭ symbol depicts a task or activity

&

In Relationship Maps – ▭ symbol depicts a department

In Basic Process Map – There is No information about the department where the task is carried out.

Block = Transfer fn
Arrow = Flow of Signal

4.1 Block Diagrams

Block diagrams are commonly applied in linear control theory to portray the relationships among components of a physical system or process. A simple example is illustrated in Figure 9. In control system applications, arrows represent the flow of signals or variables between blocks, and blocks contain transfer functions that define how the input signals are mathematically transformed into output signals. A transfer function is the ratio of the output signal to the input signal.[6] For example, the variable y in Figure 9 equals the variable x_3 multiplied by the transfer function B.

In addition to blocks and arrows, two other operations are possible in block diagrams. First, two or more separate variables can be summed algebraically using a summing node, indicated by the circle enclosing a summation symbol in Figure 9. Thus, the variable x_2 is equal to x_1 minus z. Second, any given signal can be used as an input to more than one block. In Figure 9, the variable x_3 is used as an input to blocks B and C. The operation is called a takeoff point, shown in the diagram as a large dot between blocks A and B.

In linear control theory and other physical system applications, a form of algebra (called "block diagram algebra") can be used to reduce a complex block diagram to a single block containing the input/output relationship for the entire system. Thus, a block diagram is useful for developing the input/output functions of the individual components of a control system and then analyzing the diagram to determine overall system performance.

Our applications of block diagrams in this book deal with systems that tend to be more subjective and operations oriented. In general, our systems do not lend themselves to the mathematical treatment by which physical systems can be analyzed. Nevertheless, the block diagram format is a powerful means of depicting the signal flows and interrelationships

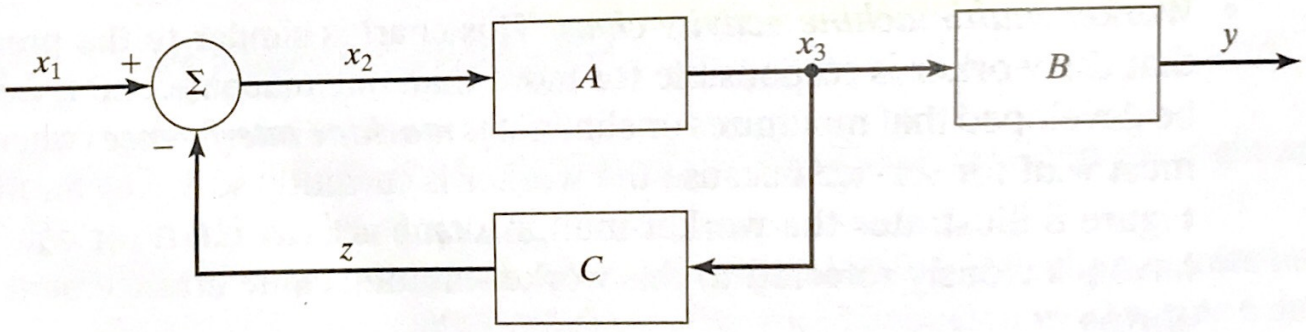


Figure 9 Block diagram of a feedback control system.

among components in complex systems that are related to work. We use them frequently throughout this text.[7] In addition, block diagrams have clearly influenced the development of other diagramming techniques in operations analysis, such as process maps.

4.2 Process Maps

Process mapping techniques have been widely applied to business processes. They are also applicable to production, logistics, and service operations. As in other diagramming and charting techniques, process maps provide a detailed picture of the process or system of interest that is helpful for communicating and understanding by those involved in the operations analysis study.

There are several forms of process maps. The basic process map is a block diagram that shows the steps in the process of interest. A ***process*** is defined as a sequence of tasks or activities that add value to one or more inputs to produce outputs. The inputs and outputs may be materials, products, information, services, or other form. The process must consist of more than one step, and the steps are linked together in a logical way. The process has a beginning point and an ending point, and its purpose in the organization is to provide something of value to its customers.

Block symbols used in the basic process map are shown in Figure 10, and an example of their arrangement in a process map is illustrated in Figure 11. The block for

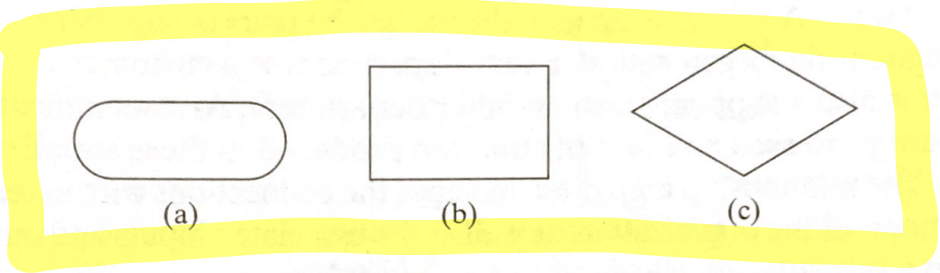


Figure 10 Symbols in the basic process map: (a) beginning/ending point of the process, (b) task or activity step, and (c) decision point. Brief labels or names are written in the various blocks to identify them.

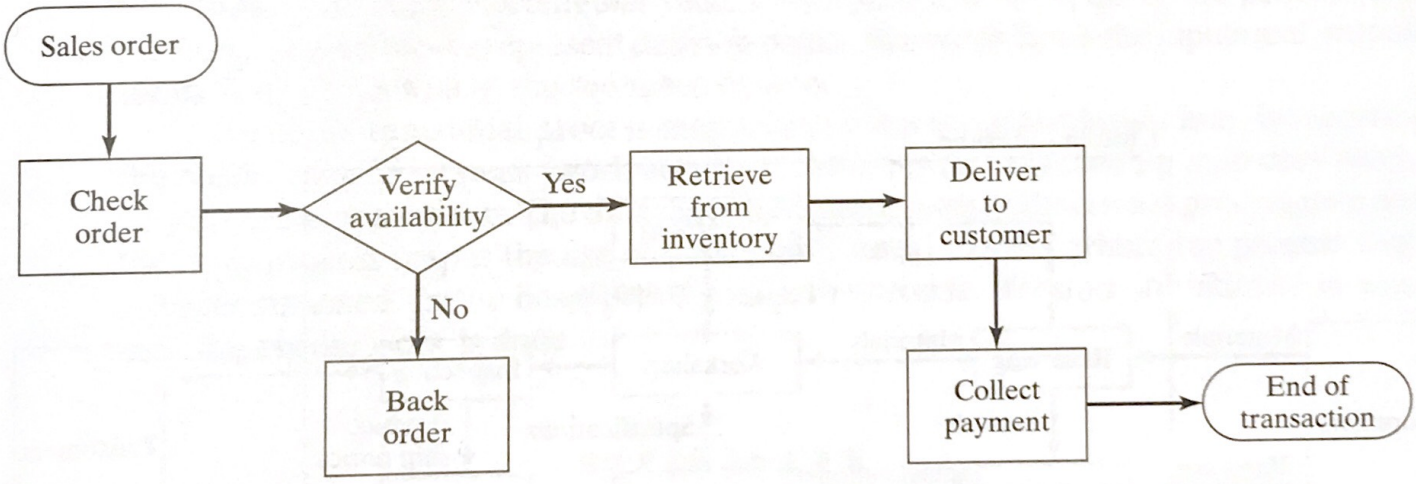


Figure 11 An example of a basic process map.

the beginning point of the process is a racetrack oval. It is the boundary between the process under study and the activities that occur outside the process under study. The same symbol and interpretation apply to the end of the process. Rectangular blocks are used to symbolize tasks or activity steps in the process. Decision points are diamond-shaped blocks. Labels in the blocks identify what each step and decision point does, as suggested in Figure 11. Arrows are used to indicate the sequence of the steps and the flow of the work.

Process mapping can be accomplished at various levels of detail. A ***high-level process map*** depicts a macroscopic view of the process of interest and includes only the most important steps in the process. Each step in the high-level map consists of tasks or work elements, which can be examined in greater detail to create a ***low-level process map*** for that step, also called a ***detailed process map***. When this is done, the blocks in the high-level map are shadowed or otherwise highlighted to indicate that the lower-level process map exists. In some cases, more than two levels of process maps may be developed for processes requiring such detail.

High level Process Map.

Alternative forms of process maps include relationship maps and cross-functional process maps [4]. A ***relationship map*** is a block diagram that shows the input-output connections among the departments or other functional components of an organization. The map consists of blocks that represent the departments and arrows to show the flow of work. A relationship map illustrates the pairs of supplier-customer relationships throughout the organization. Every department is a customer of another department, and it is also a supplier to some other department. Arrows indicate the flow of inputs that are processed and outputs that are produced in these supplier-customer relationships. The relationship map often includes the connections with external suppliers and/or customers of the organization, as well as the associated inputs and outputs. Figure 12 illustrates a relationship map for a custom workshop.

A ***cross-functional process map*** is a block diagram that shows how the steps of a process are accomplished by the various departments or other functional groups that contribute to it. As shown in Figure 13, the departments are listed in rows separated by dashed lines. This format causes the cross-functional process map to also be known as

In Relationship, Block = Department
Arrow = Flow of work.

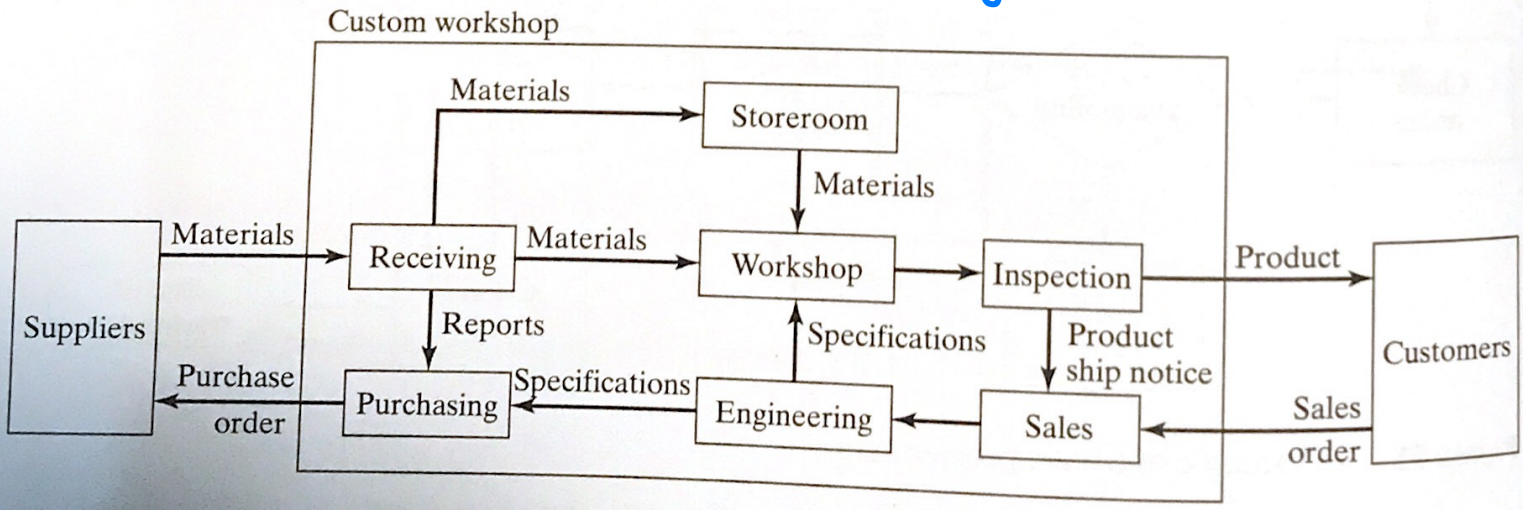


Figure 12 Relationship map for a custom workshop.

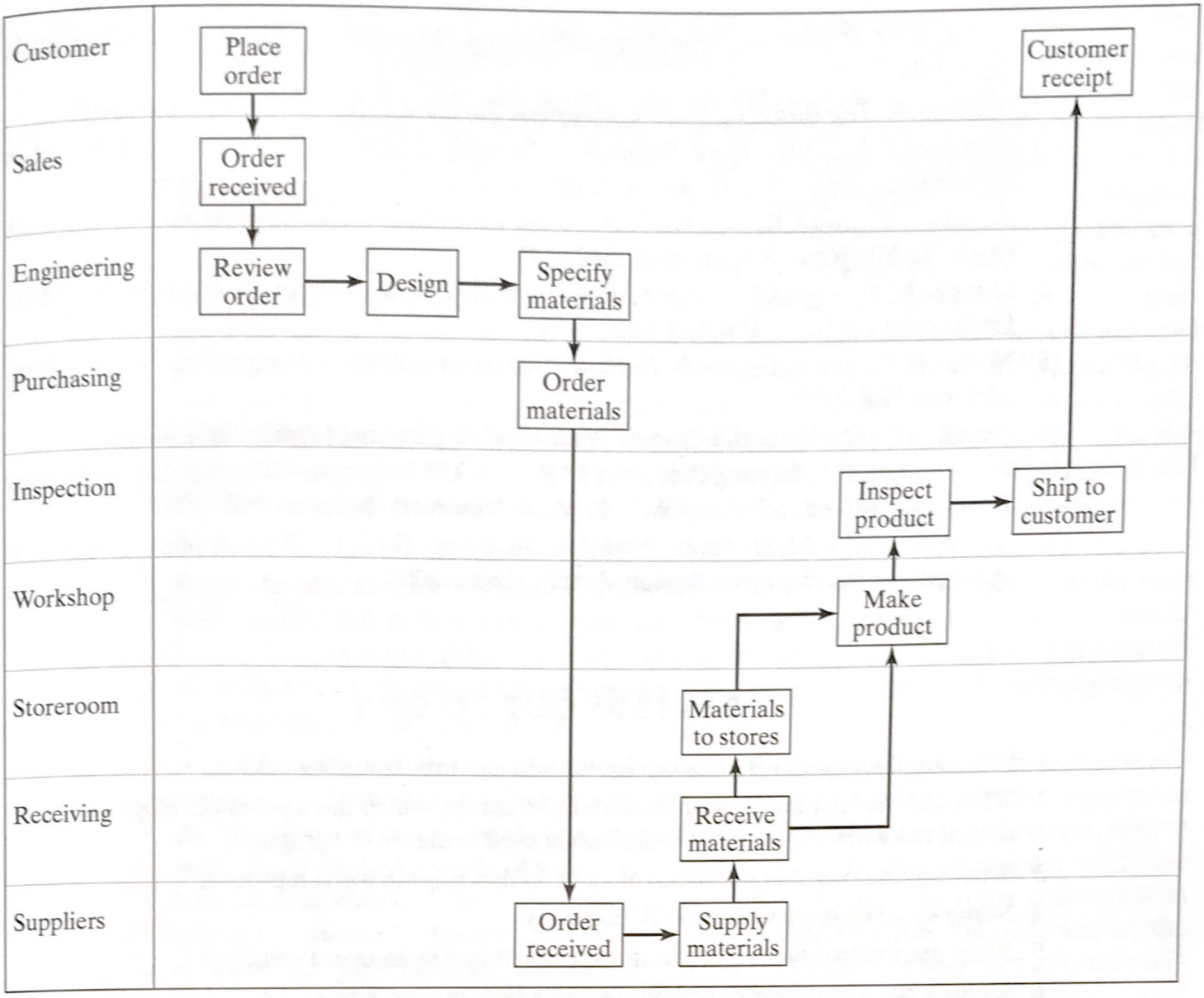


Figure 13 A cross-functional process map for a custom workshop.

the *swim-lane chart*. Rectangular blocks represent activity steps in the process, and diamond-shaped blocks represent decision points. Arrows indicate the inputs and outputs for each block, as well as the sequence of steps.

The cross-functional process map differs from the relationship map by showing the blocks as process steps (work activities), whereas the relationship map uses blocks to represent departments. The difference between the cross-functional process map and the basic process map is the use of rows (swim lanes) to show where the process steps are accomplished. In the basic process map, the process steps do not indicate in what department the work is done.

→ SIMO Charts -

- It is h/s Simultaneous Motion Chart
- It is similar to THPC, but is illustrated with "MICROMOTIONS ELEMENTS" aka BME.
- Unit of measurement of SIMO chart = 1 wink

where 1 wink = 0.1 - 0.4 seconds = $\frac{1}{100}$ mins.

Motion Study & Work Design.

- Motion Study = It involves the analysis of the basic arm, hand & body movements.

- Work Design = It involves the study of methods & motions used to perform a task.

- "Principles of Motion Economy" are used to improve the efficiency & effectiveness of manual work.

⇒ Basic Motion Elements & Work Analysis

① Therbligs :

- Therbligs includes mental & physical elements.
- Therbligs are used extensively in W.M techniques of PMTS & in particular MTM & MOST.
- "MICRO MOTION" analysis - It is used analysis the Therbligs in repetative tasks.

- OBJECTIVE'S of MICRO MOTION Analysis:

1. Eliminate ineffective Therbligs
2. Avoid Hands to Hold parts.
3. Combine Therbligs
4. Simplify the overall method
5. Reduce Time req for the Motion.

∴ Here the 17 Therbligs are -

1. TRANSPORT EMPTY = TE
   (also Reach)
2. Grasp = G
3. TRANSPORT LOADED = TL
4. Hold = H
5. Release load = RL
6. Preposition = PP
7. Position = P

8. Use = U
9. Assemble = A
10. Disassemble = DA
11. Search = SH
12. Select = ST
13. Plan = P_n
14. Inspect = I
15. Unavoidable Delay = UD
16. Avoidable Delay = AD
17. Rest = R

↳ Classification of Therbligs -

- They are classified into Effective & Ineffective Therbligs vs the degree physical & mental & delay elements.

TABLE 2 Classification of Therbligs

	Effective Therbligs	Ineffective Therbligs
Physical basic motion elements	Transport empty (TE) Grasp (G) Transport loaded (TL) Release load (RL) Use (U) Assemble (A) Disassemble (DA)	Hold (H) Preposition (PP)
Physical and mental basic motion elements	NIL	Position (P) Search (Sh) Select (St) Plan (Pn)
Mental basic elements	Inspect (I)	
Delay elements	Rest (R)	Unavoidable delay (UD) Avoidable delay (AD)

⇒ Principles of Motion Economy :-

- They are employed to MAXIMIZE the efficiency and REDUCE FATIGUE of the worker.

- Principles of Motion Economy are Categorized into 3 categories -

(1) Principles related to "USE of HUMAN BODY"

(2) Principles related to "WORK PLACE Arrangements"

(3) Principles related to "DESIGNING of TOOLING & EQUIPMENT"

⇒ (I) Principles related to use of "HUMAN BODY."

- Principles of Motion Economy are applicable for "MANUAL Work" ie either repetitive (or) non-repetitive.

- (1) Central Theme of 1st (5 principles) is the "usage of both Hands"
- (2) Next "5 principles" are based of the laws of "PHYSICS"
- (3) The last principles are to usage other body parts apart from Hands & Arms.

∴ Principles are :

1. BOTH HANDS SHOULD BE FULLY UTILIZED
2. The 2 Hands Should BEGIN & END their motions @ SAME TIME.
3. Motions of Hands should be SYMMETRICAL & SIMULTANEOUS.
4. Work Should be designed to EMPHASIZE the worker's PREFFERED HAND.
5. Worker's 2 Hands MUST never be IDLE @ the Same Time.

---

6. Method Should consist of SMOOTH CONTINUOUS CURVED MOTIONS rather than STRAIGHT-LINE MOTIONS with ABRUPT CHANGES in DIRECTION.
   - ↳ Smooth Continuous Curved motions reduce the Time lost in Accelerations & decelerations in Straight line Motions
   - ↳ Here the momentum is being conserved
7. MOMENTUM Should be used to facilitate the Task wherever POSSIBLE.
8. Method Should Take ADVANTAGE of GRAVITY & not OPPOSE it.
9. Method Should achieve a NATURAL RHYTM of the motions involved.

(10) Lowest classification of Hand & Arm motions must be used.

↳ • (1) → Finger Motions ONLY

(2) → Finger & Wrist Movements

(3) → Finger, Wrists & Forearm movement

(4) → Finger, Wrists, Forearm & Upper Arm motions

(5) → Finger, Wrists, Forearm, Upper arm & Shoulder motions.

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(11) MINIMIZE "EYE FOCUS" & "EYE TRAVEL" actions

↳ • EYE FOCUS = Occurs when the EYE must adjust to a change in viewing distance.

↳ • EYE TRAVEL = Occurs when the EYE must adjust to a LINE of SIGHT change.

(12) Method should be designed to utilize the worker's FEET & LEGS when appropriate.

⇒ (II) Principles related to "WORK PLACE ARRANGEMENTS".

• 1st 3 principles deal with "IMMEDIATE WORK AREA & NATURAL RHYTHM of work."

- The other principles cover the use of Gravity & general conditions of the workplace.

∘∘ Principles are -

(1) Tools & Materials must be located @ FIXED positions within the WORK AREA.

(2) Tools & Materials must be located CLOSE to where they are used.

(3) Tools & Materials should be placed in locations that are CONSISTENT with the sequence of WORK Elements in the work cycle.

(4) GRAVITY FEED BINS should be used to deliver SMALL Parts & FASTENERS.

(5) GRAVITY DROP Chutes should be used for COMPLETE work units when appropriate.

(6) Adequate ILLUMINATION must be provided for the workplace.

(7) A proper CHAIR should be provided for the Worker.

(III) Principles related to "DESIGN OF TOOLS & EQUIPMENT"

Principles are:

1. Work Holding devices should be DESIGNED for a Task, & worker's Hand should NEVER be used as a Work Holder.

2. HANDS should be RELIEVED of Work Elements performed by the FEET.

3. MULTIPLE Tool fns should be COMBINED into 1 Tool.

4. MULTIPLE operations should be performed "SIMULTANOUSLY" rather than "SEQUENCIALLY".
   - Eg: Pneumatically Powered Tools
     Multiple Spindle Drill Presses.
     Wave soldering

5. When FEASIBLE, an operation should be performed on MULTIPLE Parts SIMULANTOUSLY.
   - Eg: Drilling Holes on PCB.

6. Equipment Control Should be designed for operator CONVENIENCE & ERROR AVOIDANCE.

7. Hand Tools & Powered Tools should be designed for operators COMFORT & CONVENIENCE.

8. Manual ops Should be AUTOMATED wherever "ECONOMICALLY & TECHNOGICALLY" feasible

UNIT-3

(I) - Introduction to Work Measurement

(II) - Direct Time Study

(III) - Pre-Determined Motion Time Systems.

Introduction to Work Measurement.

- **Work Measurement** – It refers to the 4 Techniques that are used for the evaluation of a task in terms of time that must be allotted to a Average worker to complete that task.

- The task usually measured is "Repetative work".

Q. What is the Objective of Work Measurement?

Ans- The objective of work measurement is to determine the "STANDARD TIME."

- Work measurement is often called Time Study

Q. What is Time Study?

Ans- Time Study - It refers to all the ways in which time is investigated and analysed in a work situation.

Note - Time Study also focus on the Learning Curve phenomenon which is not included in the defn of Work Measurement.

↳ Some Information -

(1) Rating, Observed Time & Allowances are used to determine a Standard time.

(2) Any overtime is a Financial & Productive Burden.

(3) Incentives - It is an Extra Amount given over and above salary. (aka Commission)

(4) Wages - It is the amount paid equal to the amt of work achieved.

(5) Bonus - They are given provided that there is an appreciable profit for the organization.

Q. What are Time Standards & how are they determined?

Ans: Time is objective, quantifiable & people understand time.

- Standard Time (Tstd) (aka Allowed Time) = It is the amount of time that should be allowed for an average worker to process one work unit using the Standard Method & working at a normal pace.

- Time Standards are required when -
  - (i) Low Productivity
  - (ii) Repeative Work Cycles
  - (iii) Short Cycle Times.

- When done right Time Standards do the following fns -
  - Define a "Fair Day's work."
  - Determine Man Power Requirements.
  - Provide a basis for Wage Incentives.

↳ Methods of Determining Time Standards -

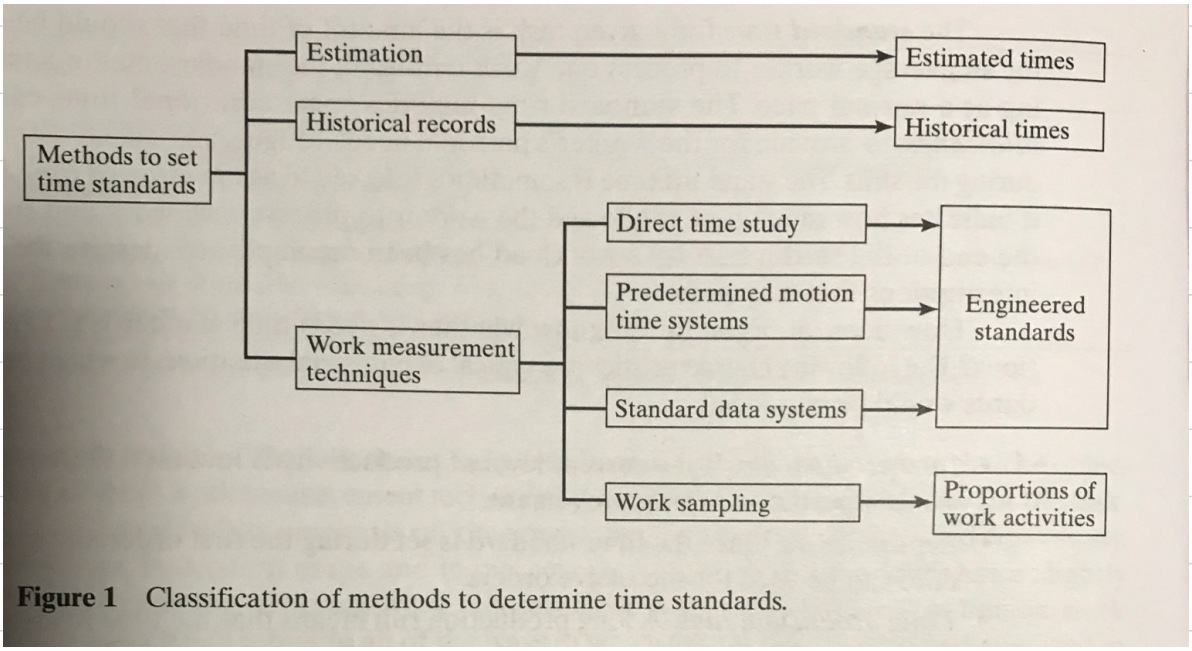


Figure 1 Classification of methods to determine time standards.

WM is to measure the amount of effort invested in terms of time.

WM is done so as to enhance the productivity in the long run.

① **Estimation** - Since this method dependents on the Estimator's Judgement, it is the least accurate of techniques to determine time standards. It is also "Guess work". It is used in "Micro Industries".

② **Historical Records** - They represent actual amt of work achieved in the past from a worker.
- However they do NOT indicate the worker's efficiency.

③ **Work Measurement Techniques** - They are more time consuming to implement but they are most accurate to determine Time Standards.

Q. What are the techniques of Work Measurement?

Ans: There are 4 techniques of WM:

1. Direct Time Study (DTS)
2. Predetermined Motion Time Study (PMTS)
3. Standard Data Systems (SDS)
4. Work Sampling.

Here DTS, PMTS, SDS are Engg. Techniques of WM.

↳ Note: Work Sampling is not reliable due to statistical errors.

Q. What is the main purpose of Work Sampling?

Ans - Work Sampling - Is mainly used to determine the proportions of time spent in various categories of work activity

⇒ 1. Direct Time Study (DTS) -

- It involves the direct observation of a Task using a stopwatch.
- Here the work elements are timed separately.

- Here the worker's pace is evaluated & hence is f/s "Performance Rating".

∴ Observed Time × Performance Rating = Normal Time.

ie $T_n = T_{obs} \times PR$

- DTS is the most common & often used techniques of WM.
- Here the Performance Rating is a "LEVELLER".

⇒ (2) Pre-determined Motion Time Study (PMTS)-

- It relies on a DATABASE of Basic Motion Elements (BME) aka / Therbligs) such as Reach, Grasp, Move etc.
- Associated with each motion element is a set of normal times.
- Here the Observations are NOT DIRECTLY made in order to avoid any DISTRACTIONS of the worker.
- Advantages of PMTS - (i) No need to evaluate Performance Rating

(ii) Can be used to determine Standard Time before the completion of the TASK.

⇒ ③ Standard Data Systems (SDS)-

- It is similar to PMTS.
- Advantage : Here Time Standards can be set even "BEFORE the activity is performed!
- A large amount of data is req to make a database for SDS.
- 1st the work element & their respective work variables are identified to make the database.

⇒ ④ Work Sampling -

- It uses Random Sampling Techniques, so that proportions of time spent can be determined.
- It actually OBJECTIVE : To determine an appropriate allowance factor for use in setting standard times.

→ Computerised Work Measurement -

- It is comprised of 2 components which are Hardware & Software.
- They are tools that aim to "Reduce the amt of Time required to set a Time Standard"

They provide the following conveniences :-

(i) Collection of accurate data.

(ii) Automated Computations

(iii) Organizing Databases.

(iv) Retrieving data from any Database.

Q. What are the Pre Requisites / Requirements for Valid Time Standards ?

→ Pre Requisites / Req. for Valid Time Standards -

Here for Valid Time Standards the Standardized factors are -

1. Standard Input Work Unit
2. Standard Output Work Unit
3. Standard Performance of Worker.
4. Standard Environment
5. Standard Method used
6. Qualified Worker is performing the task.

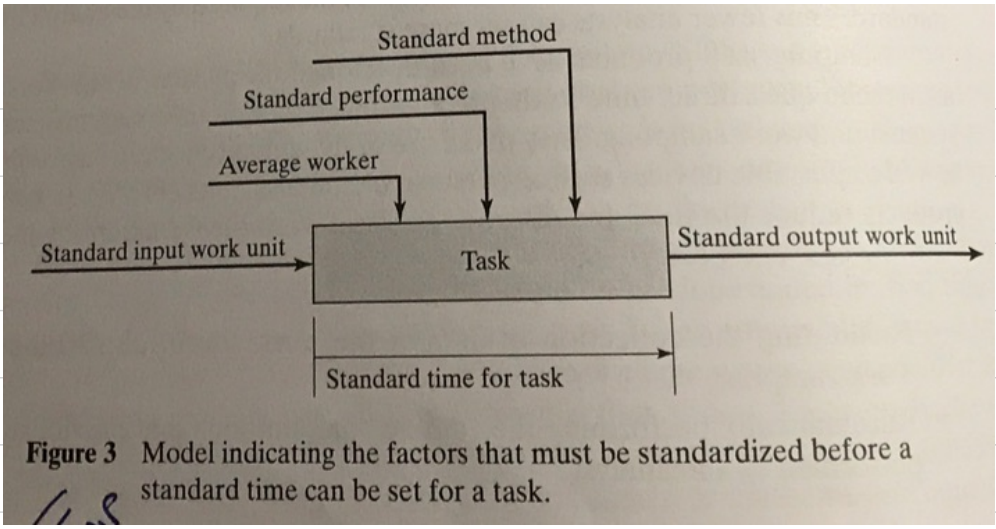


Figure 3 Model indicating the factors that must be standardized before a standard time can be set for a task.

Q. What is a Average / Qualified Worker?

Ans: Average / Qualified Worker is a person who is representative of those who perform tasks similar to the task being measured.

Q. What is Standard Performance?

Ans: It is the Pace, working at which the Standard time for the task is x% of the mean time that can be maintained throughout an entire work shift without harmful effects to the health of a Qualified Worker.

Eg - ① Walking @ 4.8 Kmph on Flat Grounds.

② Dealing 4 Hands of Cards in 30 seconds.

Normal Performance = Standard Performance.

Q. What is Normal Time?

Ans: When an individual work cycle is performed at 100% performance, the time taken is k/s Normal Time.

Standard Time is given by the eqn -

$$T_{std} = T_n (1 + Apfd)$$

NOTE: It is a common Industrial Practice to define Standard Performance as a pace that can be readily achieved by the majority of workers.

Q. What is Say Work Pace?

Ans: A multiplier of "x" means that an average/qualified worker should be able to work at a pace that is "$(x \times 100)$%" of the defined Standard pace.

→ Standard Method -

Def: "It is the procedure that has been determined to be the optimum method for processing a work unit."

- It satisfies the "One best Method Principle"
- Standard Method is that procedure that is - safest, quickest, most productive & least stressful to the worker.

Q. What are the details to be recorded by the Standard Method?

Ans:
1. Procedure used by the worker.
2. Tools used.
3. Equipment used
4. Workplace layout
5. Working conditions.

∴ Good documentation is a BURDEN to prepare & a blessing to practitioners.

→ Standard I/P & O/P work units -

- Starting & completed work units are usually easy to define in production.
- In service work, the I/P & O/P work units are usually more difficult to define.
- But service work is done by a Standard Method which in turn have some Standard Time.

→ Work Hierarchy -

- Depending on the level of work an appropriate method of work measurement can be selected.
- As per the pyramidal structure of work -

WKT- Basic Motion Elements (BME), here Therbligs are used to denote BME & its unit of measurement is (TMU).

∴ Work Element = Σ (BME)

Task = Σ (Work Elements)

Job = Σ (Tasks)

Q. What are the appropriate techniques of work measurement for BME, Work Elements & Tasks?

Ans -

For BME : PMTS

For Work Elements : DTS / SDS

For Tasks : Work Sampling.

- For BME, the Standard Time can be set even before the activity is performed.

⇒ Allowances in Time Standards -

- The purpose of an Allowance factor is to compensate for the lost time by providing a small increasement of "Allowed Time" in each cycle.

Q. How do companies decide the allowance factor for its workers?

Ans: They use 2 basic approaches -

(a) Scheduled Break Periods during the shift

(b) a PFD allowance added to the normal time.

Q. What are Scheduled Break Periods?

Ans: They are preplanned Periods set aside during the shift as a break time from work.

- In many companies Lunch & Dinner breaks are under this category.
- Typical duration of these breaks is 5 to 15 mins.
- The timings of the breaks are so planned that they divide the Work Day into equal periods.

Q. What is an PFD allowance ?

Sol: • An approximate value of personal, fatigue & delay time is determined & is then used to evaluate the standard Time from the Normal Time.

• It can be provided to compensate the work related & non-work related activities.

• Usually Bathroom, Smoke, Phone Breaks are PFD's.

P = Personal Time : • Typically it is 5%

• Personal allowance will be high provided the work environment is very Hot.

• Value of Personal Allowances is LOWER for favourable working conditions.

F = Fatigue Time - • It is also Rest Allowance.

• It is used to compensate the worker for time that is taken to overcome the fatigues & stresses, due to work.

• Allowance of Fatigue is determined by a comb" of - (a) Rest Pauses.

(b) Negotiation b/w Labour & Management.

- The Rest formulae provide a Scientific Basis
- Practically Fatigue / Rest allowances are a result of Negotiations!
- For Light & Medium weight of Work the allowance is 5%
- For Heavy weight of Work the allowance is 20% or more.

Q. What are the factors that cause fatigue?

Sol: Factors are -

(1) Physical factors :
- Standing
- Abnormal Body Position
- Use of force

(2) Mental & Cognative Factors -
- Mental Strain
- Eyestrain
- Monotony

(3) Environmental & Work factors -
- Poor Lighting
- Noise
- Heat
- Fumes.

D = Delay Time: • They are unavoidable interruptions at random times during the day.

- It is reserved for delays that are Ultimately the responsibility of management.
- Determining Delay Allowances is more "COMPLEX" than determining allowances of fatigue / personal.

• Here the Delay Time is evaluated by-

① Intensive observations over several days.

② Work Sampling.

Q. Why is Work Sampling usually preffered?

Ans: Reasons are -

(i) Work Sampling can be conducted over a long period of time.

(ii) Samples from different facilities can be collected as the "Observer is not limited to one location"

(iii) Does not require continuous observations

(iv) Statistical measures can be used to verify the accuracy of the results.

Q. What are the other types of allowances?

Sol: These other allowances are added to the Apfd allowance & are NOT used as a Substitute for Apfd.

① Contingency Allowance -

- These allowances are only provided because of some kind of problem encountered with the TASK or the Production Equipment.
- These allowance should NOT be greater than 5%.
- It may be required on a Temporary Basis

② Policy Allowance -

- They are INTENDED to cover Special Work Situations that are associated to "Wages Incentives System"

Q. What are the different types of POLICY allowances -

Sol:
① Machine Allowance
② Training Allowance
③ Leaving Allowance.

Q. What are Machine Allowances -

Sol: It is the allowance that is added to the machine - paced portion of a work gen.

ie $T_{std} = T_n(1 + A_{pfd}) + T_m(1 + A_m)$ ||

Q. What are the Training allowance?

Ans: It is the "allowances" worker whose main responsibility is to teach the other workers.

Q. What is Learning Allowance?

sol: It is the allowance provided for NEW workers.

- Policy Allowances are over & above & exist due to the companies policies.
- Allowances are given to maintain the Standard Pace.
- PFD allowances takes into account only the amount of work achieved.
- Incentives are directly related to the Standard Time.

⟹ Accuracy, Precision & Application Speed Ratio:

Measurement : It is a procedure to quantify the unknown quantity with respect to a Standard.

All measurement systems are based on 7 basic physical quantities.

ie ① Length

② Mass

③ Time

④ Electrical Current

⑤ Temperature

⑥ Luminous Intensity

⑦ Matter.

3 Important attributes of the Measuring System are -

(i) Accuracy

(ii) Precision

(iii) Speed of Response.

(1) **Accuracy** - Def: It is the degree of closeness to which the measured value approaches the True Value.

- Accurate measurements are FREE from "Systematic Error" (ie +ve or -ve deviations from True value).

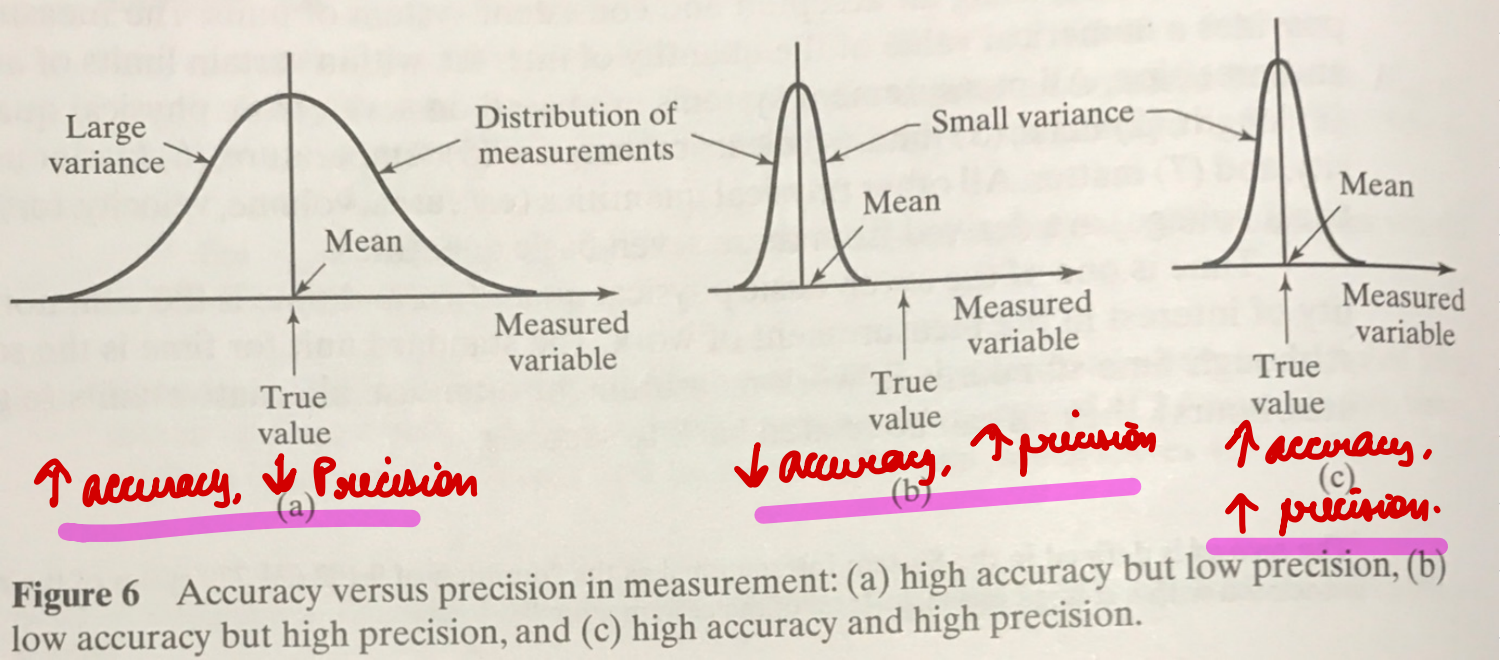


Figure 6 Accuracy versus precision in measurement: (a) high accuracy but low precision, (b) low accuracy but high precision, and (c) high accuracy and high precision.

(2) **Precision**: Def: It refers to the Repeatability of the measuring system.

- Precise Measurements are FREE from "RANDOM ERROR".

Q. "Is accuracy in Work Measurement Elusive?" Why?

Ans: It is Elusive because of the following reasons:-

1. True Value of Task Time is UNKNOWN.
2. There is No FIXED standard for comparision during measurement.

(3) Here the performance is a major variable.

(4) The normal time is a product of the observed time & the performance rating.

- However the precision of the Standard Time is determined with excellent reliability.

(3) Consistency : Def : It refers to the variations in a measured quantity between different measuring personnell.

∴ Good Consistency ⇒ Small Variations

Q. Discuss the techniques of WM & methods to define Time Standards with respect to Accuracy

Ans : (1) Estimation : Poorly Accurate.

(2) Historical Records : Fairly Accurate.

(3) Work Measurement Techniques : (a) Work Sampling : Fairly Accurate

(b) PMTS : Fairly Accurate

(c) DTS : Good Accuracy

(d) SDS : Very Good Accuracy.

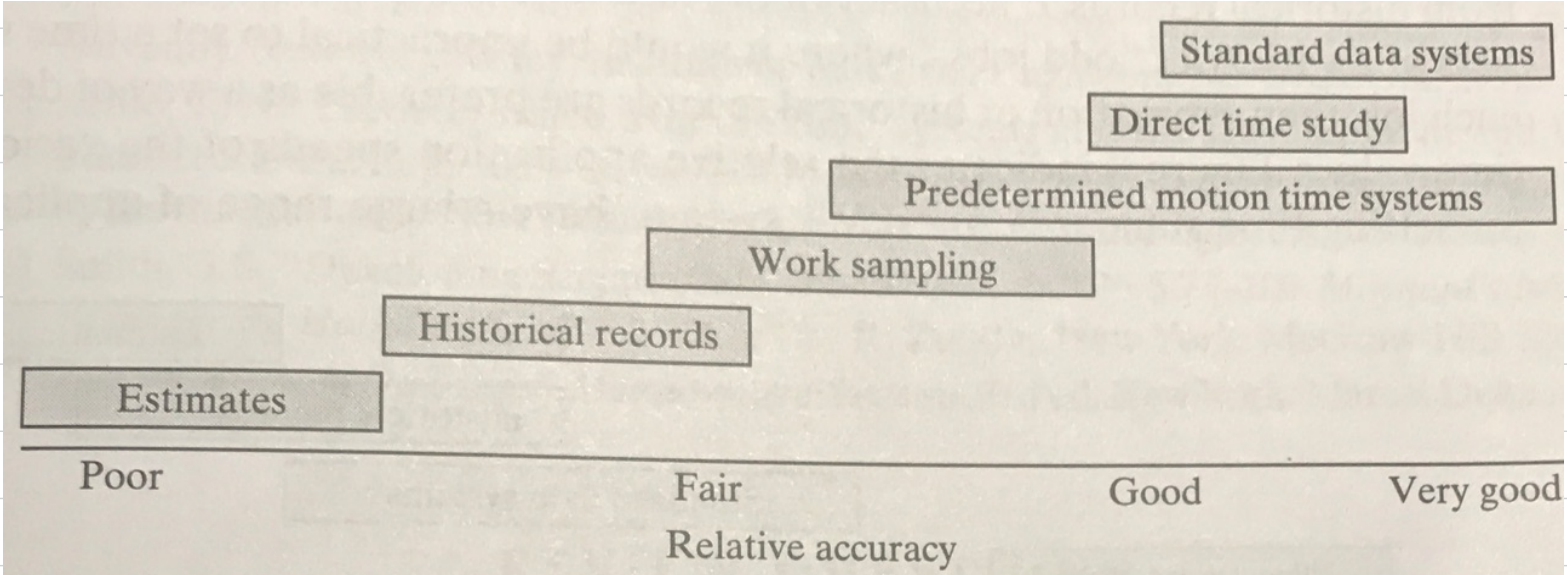


Figure 7 Methods of determining time standards ranked by relative accuracy.

(4) Application Speed Ratio: (aka Speed of Response)

Def: "It is the lag of time b/w the change in the quantity & the indication of that change by the instrument."

Here Speed of Response is the Ratio of:

"Time req to set the Standard Time to the actual Standard Time"

- There is an INVERSE CO-RELATION b/w Application Speed Ratio & Accuracy.

Q. Discuss the techniques of Work Measurement & methods to set Time Standards with respect to "Application Speed Ratio".

Ans: (1) Work measurement Techniques : (a) Work Sampling: Omitted from Chart

(b) PMTS : Slow Speed

(c) DTS : Slow Speed

(d) SDS : Medium Speed

(2) Historical Records : Medium Speed.

(3) Estimations : High Speed.

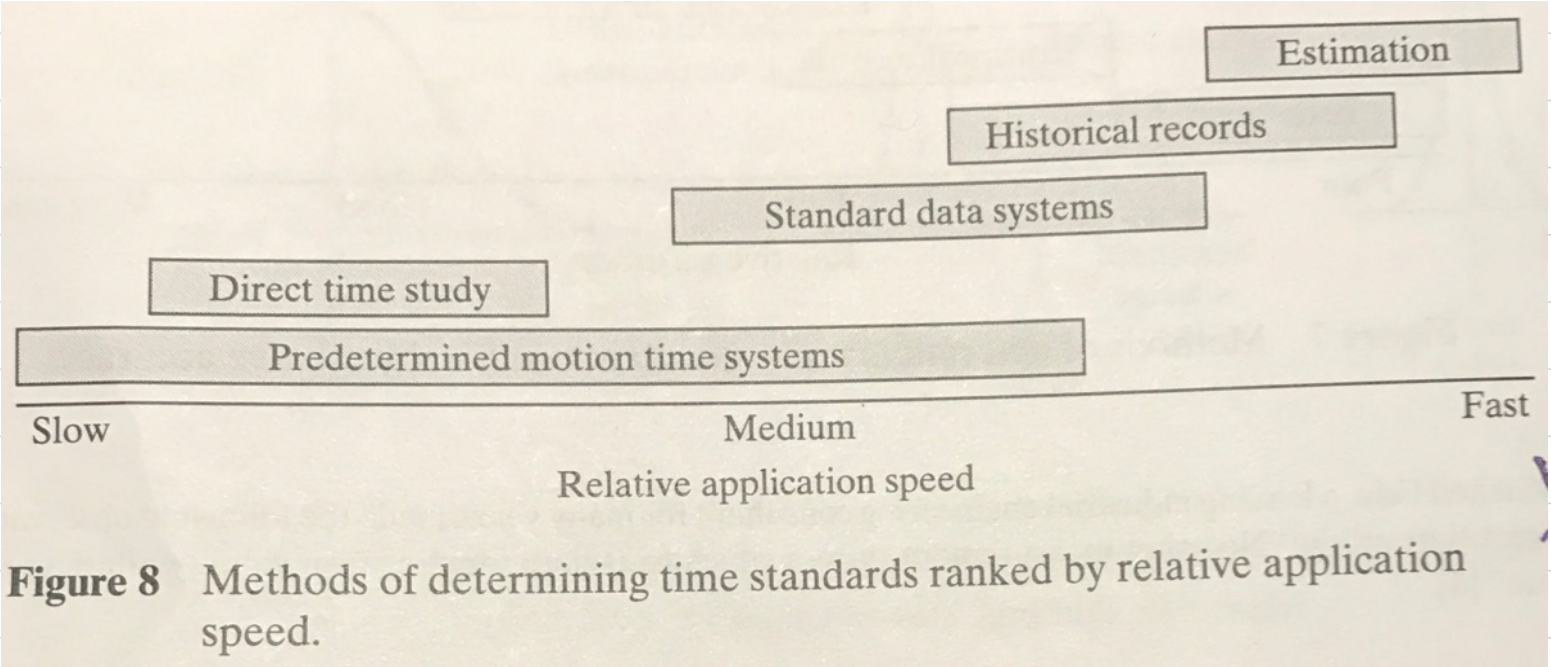


Figure 8 Methods of determining time standards ranked by relative application speed.

(II) Direct Time Study (DTS)

- Direct Time Study is aka "Stopwatch Time Study".
- It involves the DIRECT & CONTINOUS observation of a Task using a Stopwatch.
- DTS was the 1st techniques of WM to be used & it was developed in 1883.
- DTS is preffered for - (1) Routine Office Work
  (2) Repetitive in Batch Prodn.
  (3) Repetitive in Mass Prodn.
- Limitation of DTS -
  "The Standard Time cannot be set before the start of production."
- DTS involves Elemental Rating & Elemental timing.
- The Performance Rating is the "Levelling factor" for the variations in the Normal Time!

↳ Direct Time Study Procedure -

- The procedure to set Standard Time through Direct Time Study consists of 5 steps.

Step (1) : "Define & document the Standard Method"

Step (2) : "Divide the Task into Work Elements".

Step (3) : "Time the Work Element & obtain the Observed Time".

Step (4) : "Do performance rating against the Standard performance".

Step (5) : "Apply the allowances to the Normal Time & hence find the Standard Time".

↳ Detailed Procedure to set Time Standards by DTS -

(1) Define & Document the Standard Method:

- Initially a Methods Study is undertaken to ensure the methods study follows the "ONE - BEST METHOD" principle.
- State explicitly any irregular work cycles & their frequency of occurance.

- Once a Standard method is develop, it must impossible to make further improvements
- Throughly document the Standard method.

Q. Why is it important to document the Standard Method?

Ans: (1) Batch Prod^n - • Here the repeation of work is guaranteed for batches.

• Documentation for Batch Prod^n helps to avoid a situation of "Reinventing the wheel"

(2) Methods Improvement - • Labour - Mgnt by operator contracts only allows for "RETIMING".

(3) Disputes about the Method - • Documentation of Standard method can be used to settle any disputes with regards to the "Standard Method"

(4) Data for SDS - Data obtained by a detailed DTS can be used in developing a database for SDS.

(II) Division of Task into Work Elements -

- Work Elements - It is a combination of "BME's".

Q. What are the Guidelines to define work Elements in DTS?

Ans - (1) Work Element must consist of "Logical group of Motion elements".

(2) Begining pt of 1 element must be the "END pt" of the other preceding element".

(3) Each Element should have a "Readily Identifiable End pt".

(4) Work Elements should "NOT be too Long"

(5) Work Elements should "NOT be too short".

(6) Irregular work Elements must be "identified & Distinguish from Regular work Elements".

(7) Separate "Manual Elements from M/c elements"

(8) Separate "Internal Work Elements from External Work Elements".

(III) Time the Work Elements –

- Each Element should be timed over "Several cycles to obtain a reliable Average".
- Appropriate no. of cycles can be estimated by Statistical Evaluation.
- There are 2 principle methods of using a Stopwatch –

① Snapback Timing – Here the watch is started @ the beginning of every work element by snapping it to zero at the end of the previous element.

② Continuous Timing – Here the watch is zeroed @ the beginning of the 1st cycle & is allowed to run continuously throughout the duration of the Study.

Q. What are the advantages of Snapback method?

Ans: There are 2 advantages:

(1) Variation in Element Times is readily observed.

(2) No progressive subtraction is necessary.

Q. What are the advantages of Cumulative method?

Ans: (1) Here since the clock is running continuously, mistakes are omitted.

(2) Regular & Irregular work elements can be distinguished.

(3) Not much manipulation of the stopwatch is required.

IV Rating of Worker's Performance.

- When observing the time data, the analyst must simultaneously assess the performance relative to the def^n of the standard performance.
- Performance is aka - 1) Pace
  2) Speed
  3) Effort
  4) Tempo.
- Standard Performance = 100% Rating
- ∴ Normal / Normalized / Base Time = Obs Time × Rating
- Rating is the MOST DIFFICULT & CONTROVERSIAL step in STS.
- Workers wish to be rated @ a higher PR thus increasing the standard time & it benefits them if they are paid on a incentive wage plan.

(V) Apply Allowance & Compute Standard Time -

ie $T_{std} = T_n (1 + Aptd)$ for Pure Manual Element

- The allowance factor represents an Avg for the type of work, equipment & conditions of the work.
- For a Machine Element & Manual Element -

$$T_{std} = T_{nw}(1 + Aptd) + T_m(1 + Am)$$

Example 2 Determining a Standard Time for a Task That Includes a Machine Cycle

The snapback timing method was used in a direct time study of a task that includes a machine cycle. Elements *a*, *b*, *c*, and *d* are performed by the operator, and element *m* is a machine semiautomatic cycle. Element *b* is an internal element performed simultaneously with element *m*, and element *d* is an irregular element performed once every 15 cycles. Observed times and performance ratings are given in the table below. The PFD allowance factor is 15%, and the machine allowance is 20%. Determine (a) the normal time and (b) the standard time for the work cycle.

Worker element	*a*	*b*	*c*	*d*
Observed time, manual	0.22 min	0.65 min	0.47 min	0.75 min
Performance rating	100%	80%	100%	100%
Machine element		*m*		
Observed time, machine	(idle)	1.56 min	(idle)	(idle)

Solution **(a)** The normal time must take account of which element, *b* or *m*, has the larger value. Also, element *d* must be prorated across 15 cycles.

$$T_n = 0.22(1.00) + \text{Max}\{0.65(0.80), 1.56\} + 0.47(1.00) + 0.75(1.00)/15$$
$$= 0.22 + 1.56 + 0.47 + 0.05 = 2.30 \text{ min}$$

(b) The same comparison between elements *b* and *m* must be made in computing the standard time.

$$T_{std} = (0.22 + 0.47 + 0.05)(1 + 0.15) + \text{Max}\{0.52(1 + 0.15), 1.56(1 + 0.20)\}$$
$$= 0.85 + 1.87 = 2.72 \text{ min}$$

■

Q. How to decide on the no. of cycles to be record-ed for a through DTS & Time Standard??

Ans: The dilemma of the no. of cycles to be observed arises due to "Statistical Variations" in the no. of times of respective work elements from 1 cycles to the next.

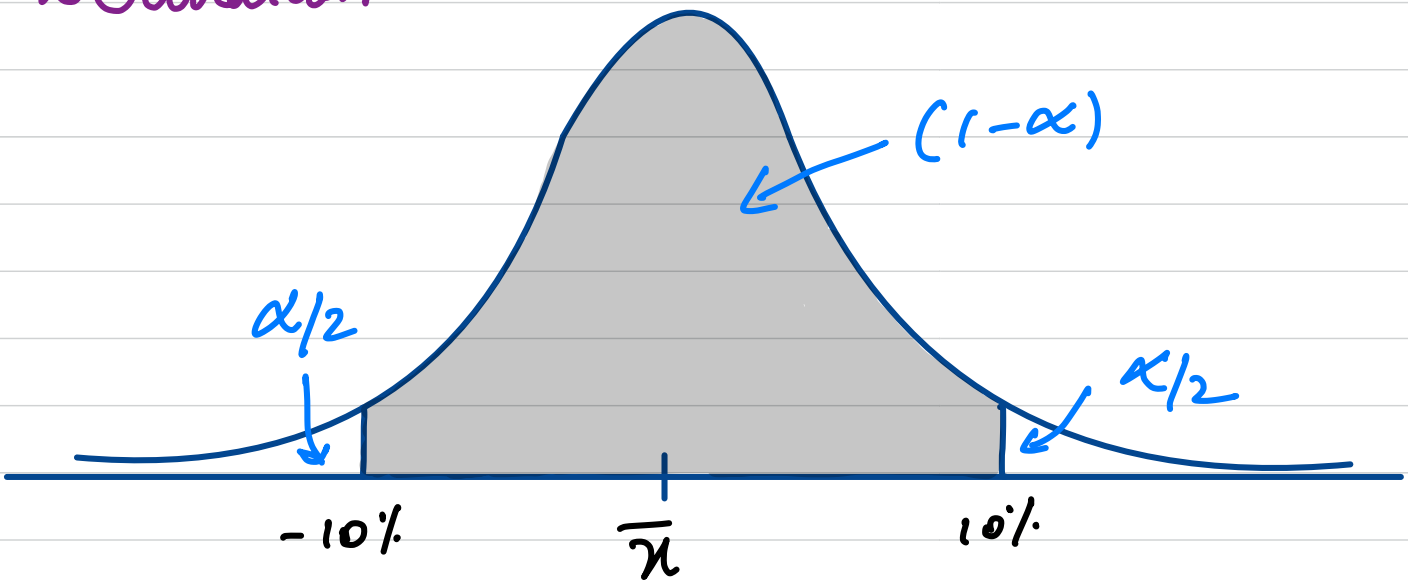
Q. Why does the work Elemental Time vary b/w cycles?

Ans: Variations occur due to -

1. Variations in Hand & body Motions
2. Variations in Placement & Locations
3. Variations in the Quality of Starting work units
4. Mistakes
5. Errors in Timing Methodology
6. Variation in Worker's Pace.

Q. Derive an expression to evaluate the no. of cycles to be observed?

↳ Derivation -



- Focusing on the Largest / Most Imp. Element & call its corresponding Elemental Term as Te.
- Confidence Interval : It is an Interval when the confidence is said to lies in.
- Average value = $\bar{x}$.
- Shaded Portion represents the probability that the True Value Te lies b/w the given Confidence Interval.

ie $$\text{Te lies within } \left(\bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = (1-\alpha)$$

Here $Z_{\alpha/2}$ = Standard Normal Variate

S = Std. Deviation, n = Sample Size, $(1-\alpha)$ = Confi Level.